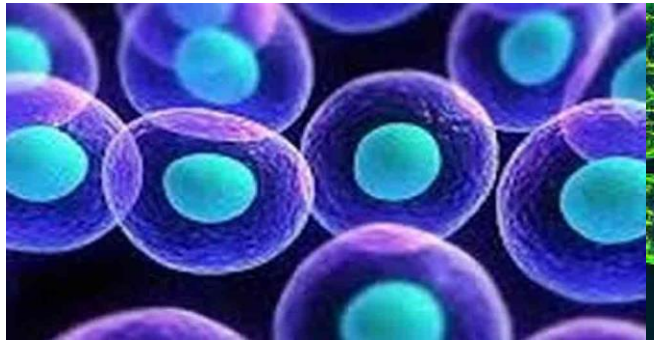
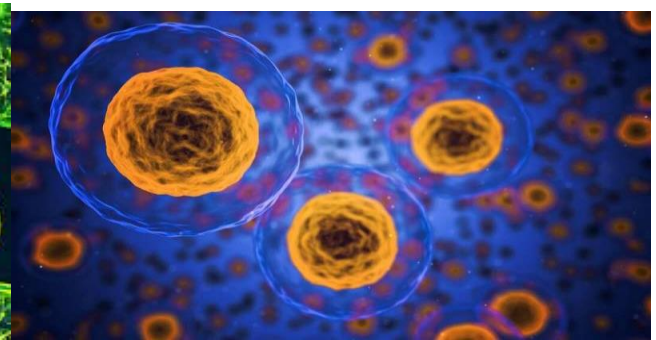
Cell structure



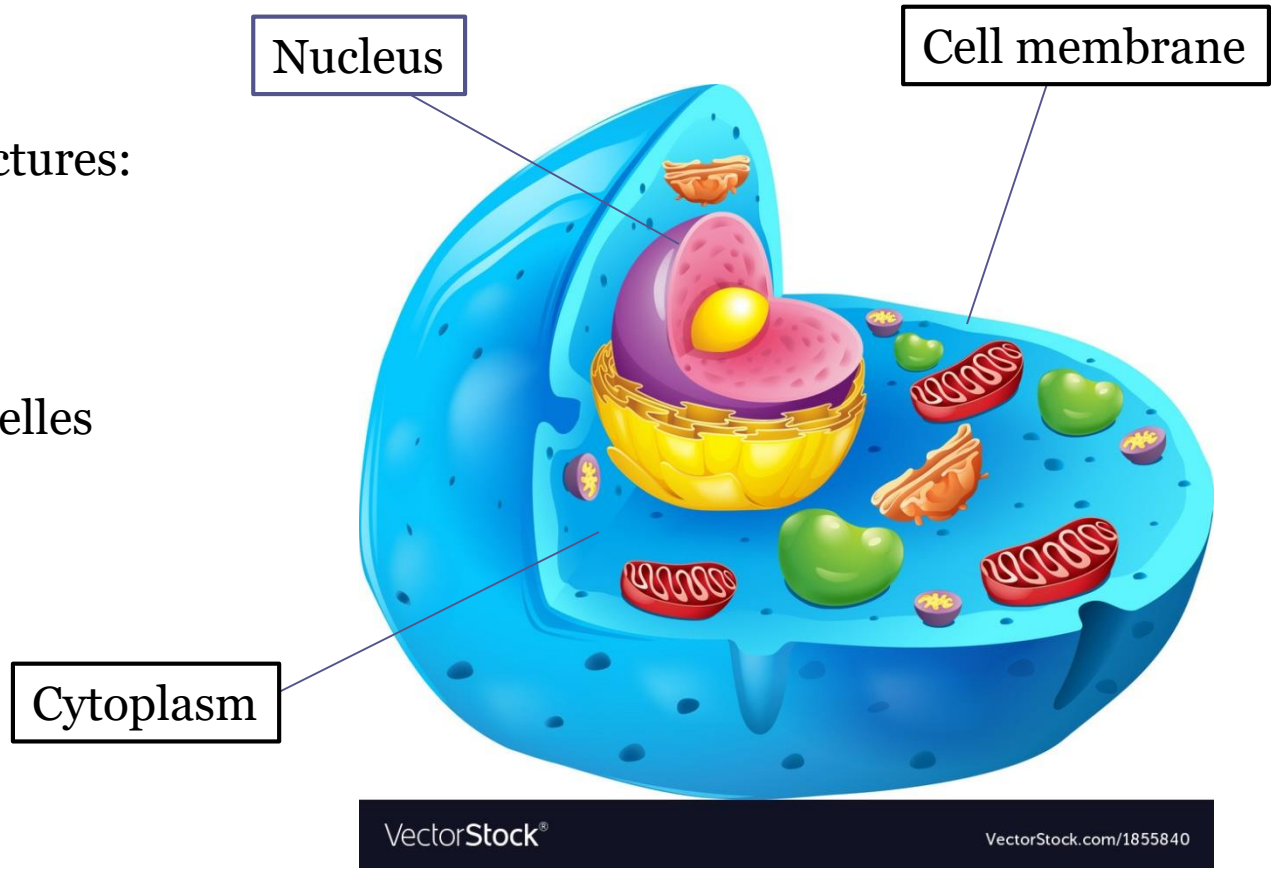




Eukaryotic Cell

Contain 3 basic cell structures:

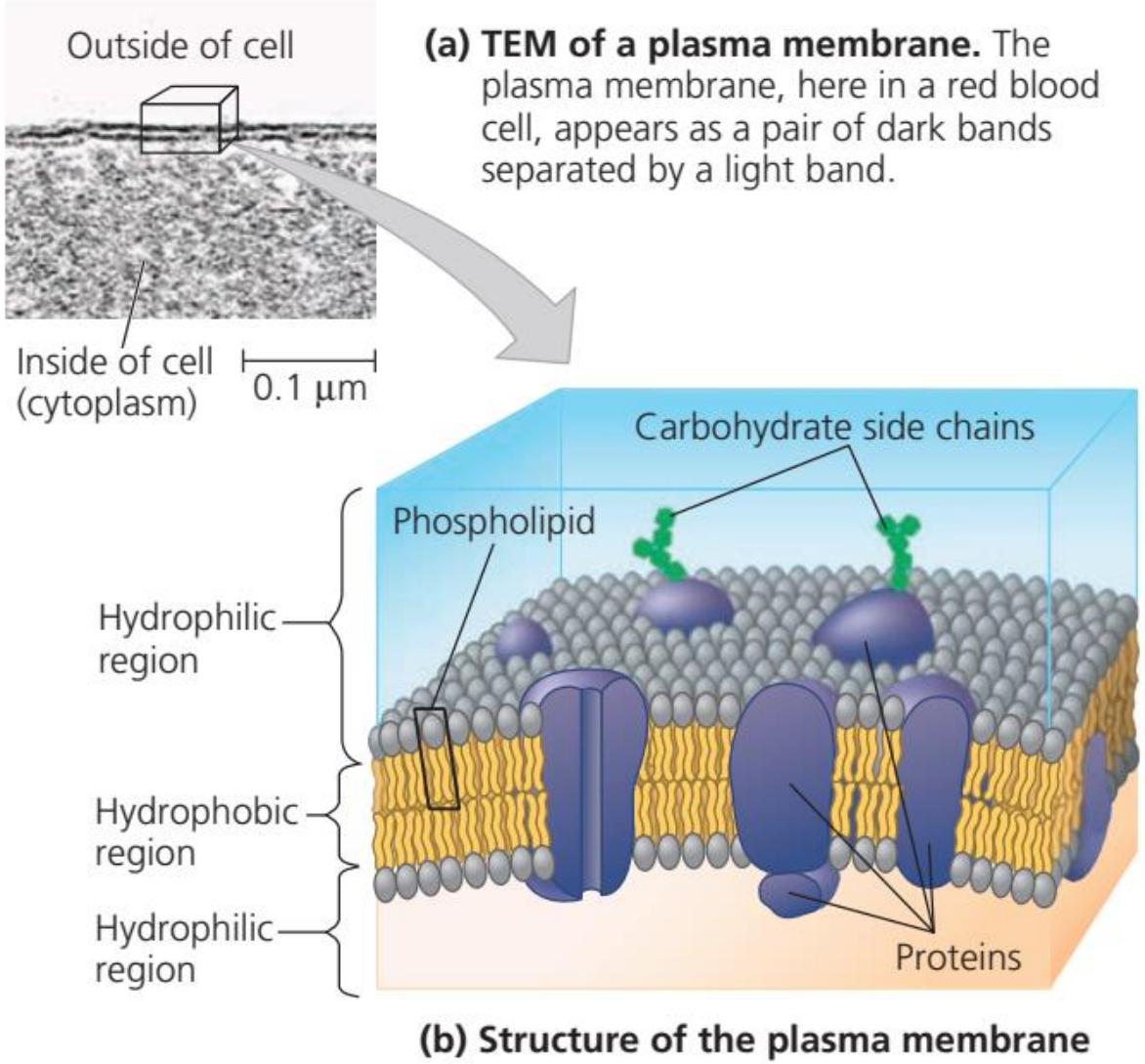
1. Cell Membrane
2. Nucleus
3. Cytoplasm with organelles



The plasma membrane

- Surrounds outside of all cells
- Living layer: it has systems for signaling between the interior of the cell and the external environment

- Composed of:
  + Double layer of phospholipids
  + Protein
  + Carbohydrates and lipids



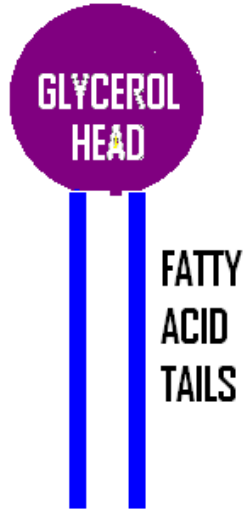
(a) TEM of a plasma membrane. The plasma membrane, here in a red blood cell, appears as a pair of dark bands separated by a light band.

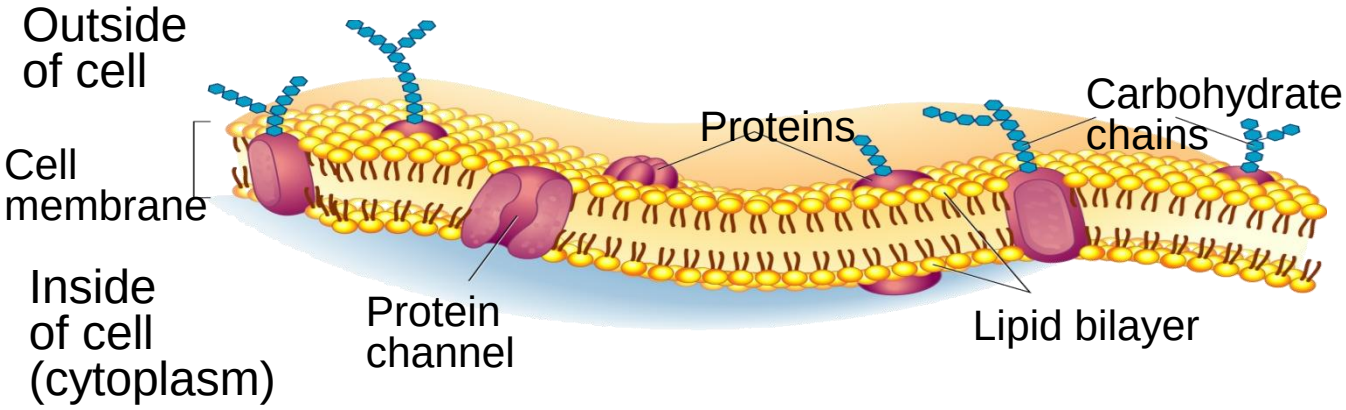
(b) Structure of the plasma membrane

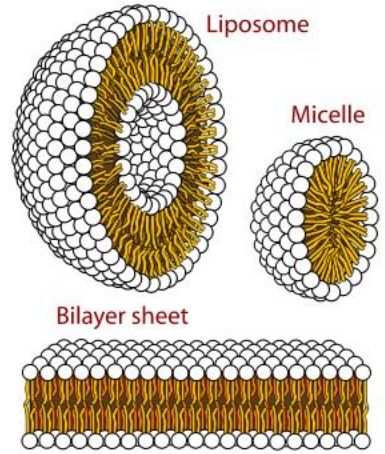
Plasma Membrane structure

Phospholipids

- Heads contain glycerol & phosphate and are hydrophilic (attract water)
- Tails are made of fatty acids and are hydrophobic (repel water)
- Make up a bilayer where tails point inward toward each other

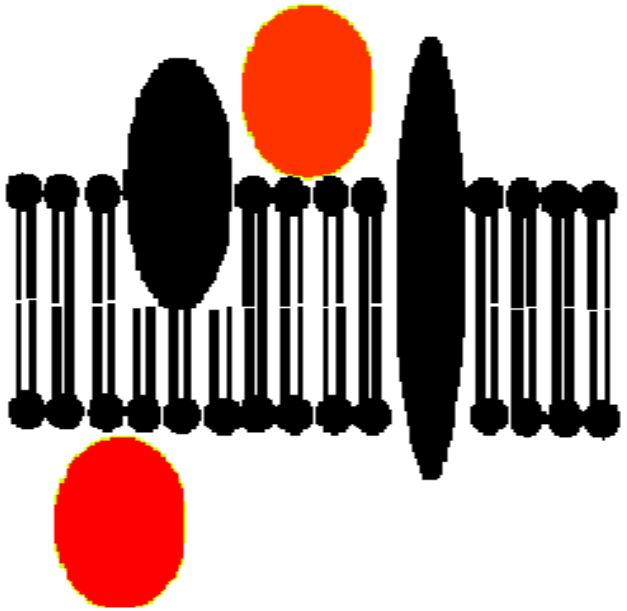






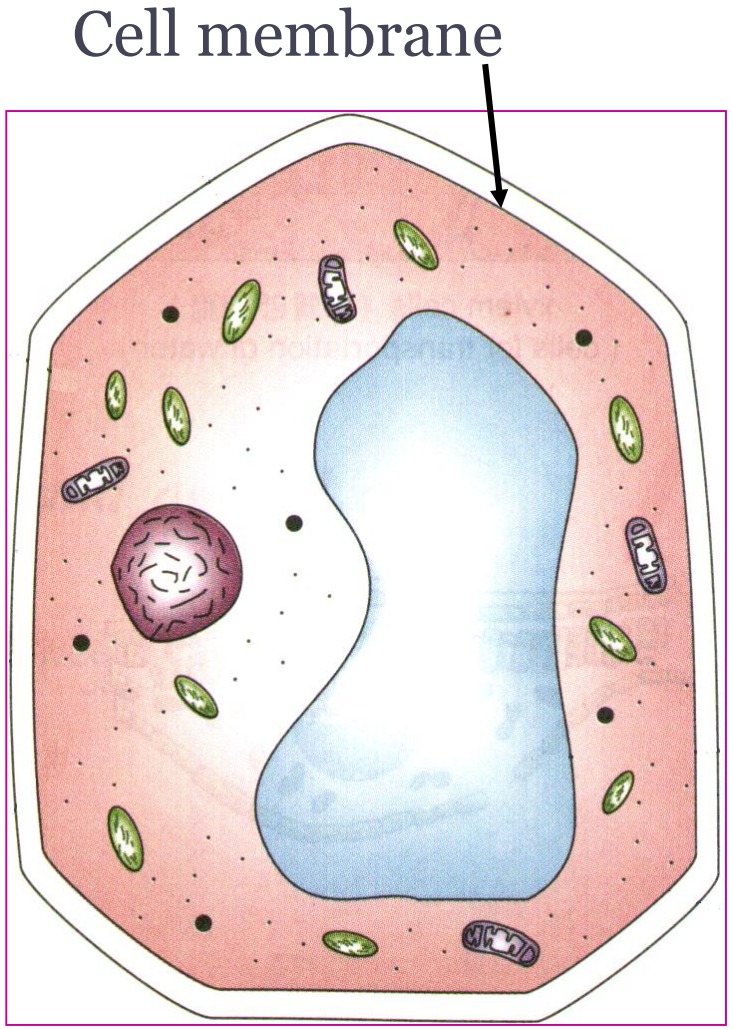
Cell membrane Proteins

- Proteins help move large molecules or aid in cell recognition
- Peripheral proteins are attached on the surface (inner or outer)
- Integral proteins are embedded completely through the membrane



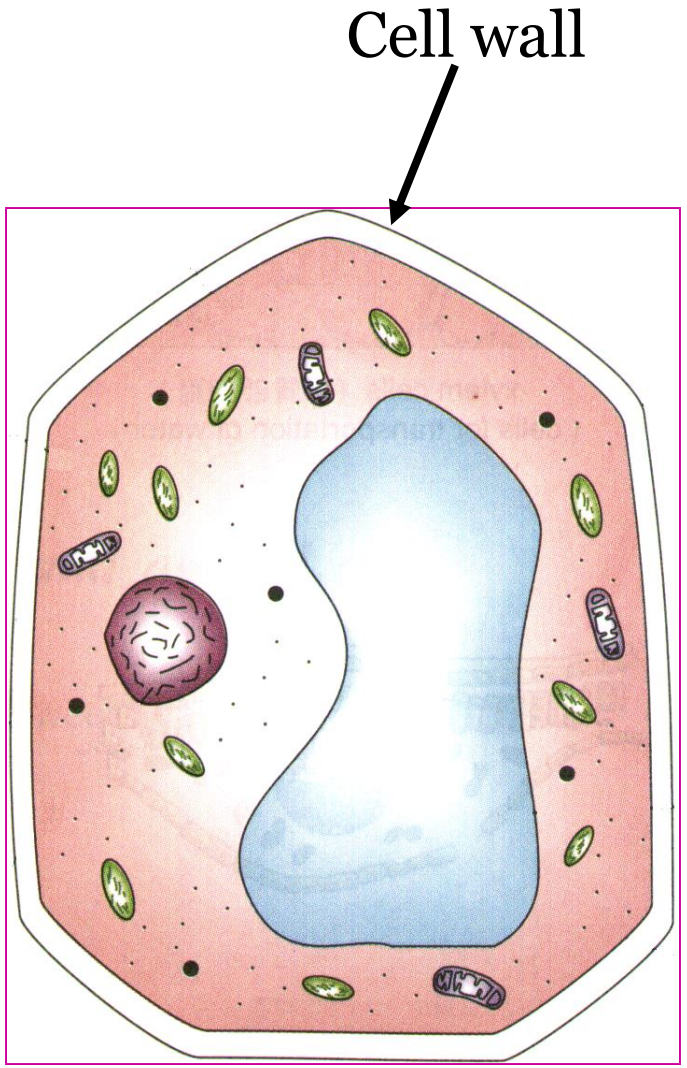
Cell membrane of plant

- Lies immediately against the cell wall in plant cells
- Pushes out against the cell wall to maintain cell shape



Cell wall

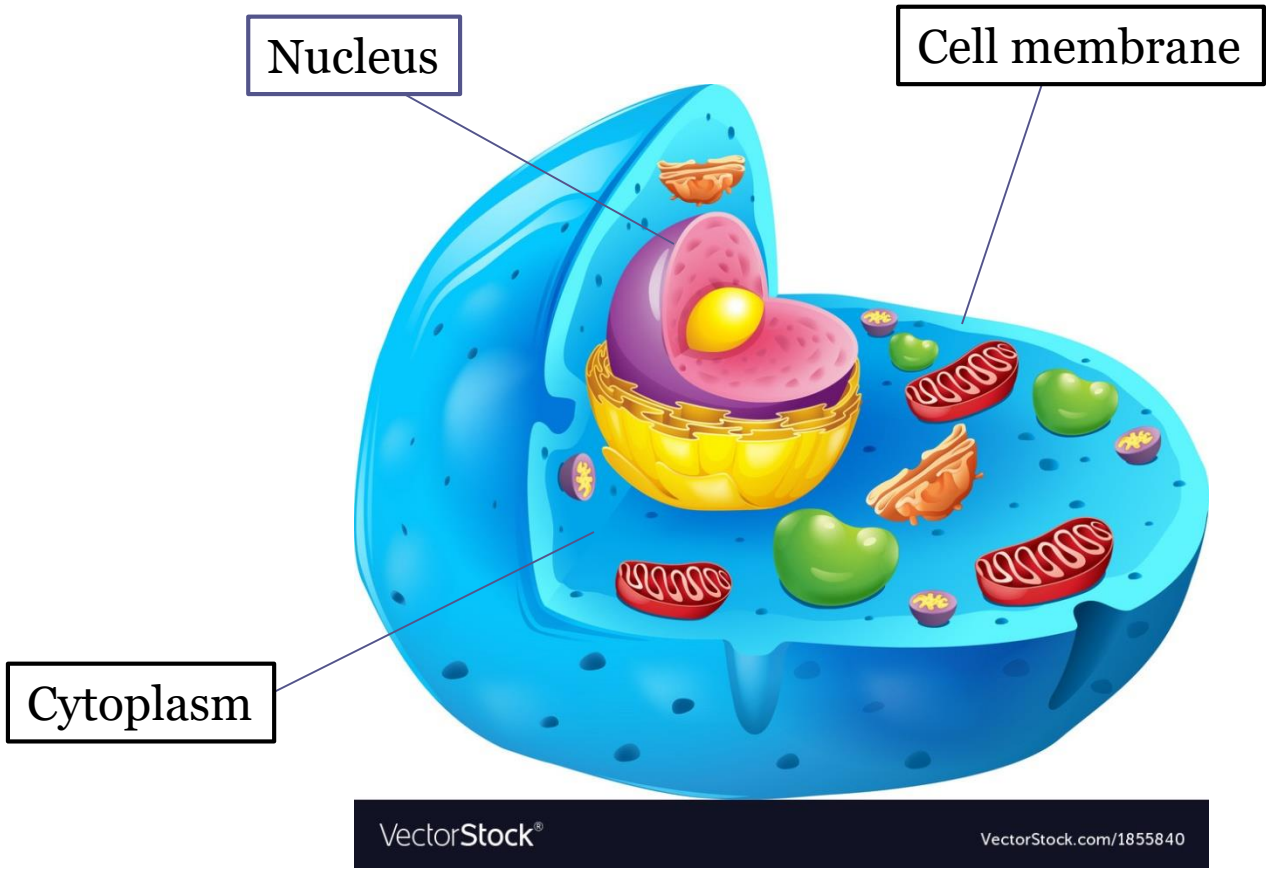
- Found outside of the cell membrane
- Nonliving layer
- Important functions in a cell include protection, structure, and support
- Found in plants, fungi, & bacteria



Eukaryotic Cell

Contain 3 basic cell structures:

1. Cell Membrane
2. Nucleus
3. Cytoplasm with organelles

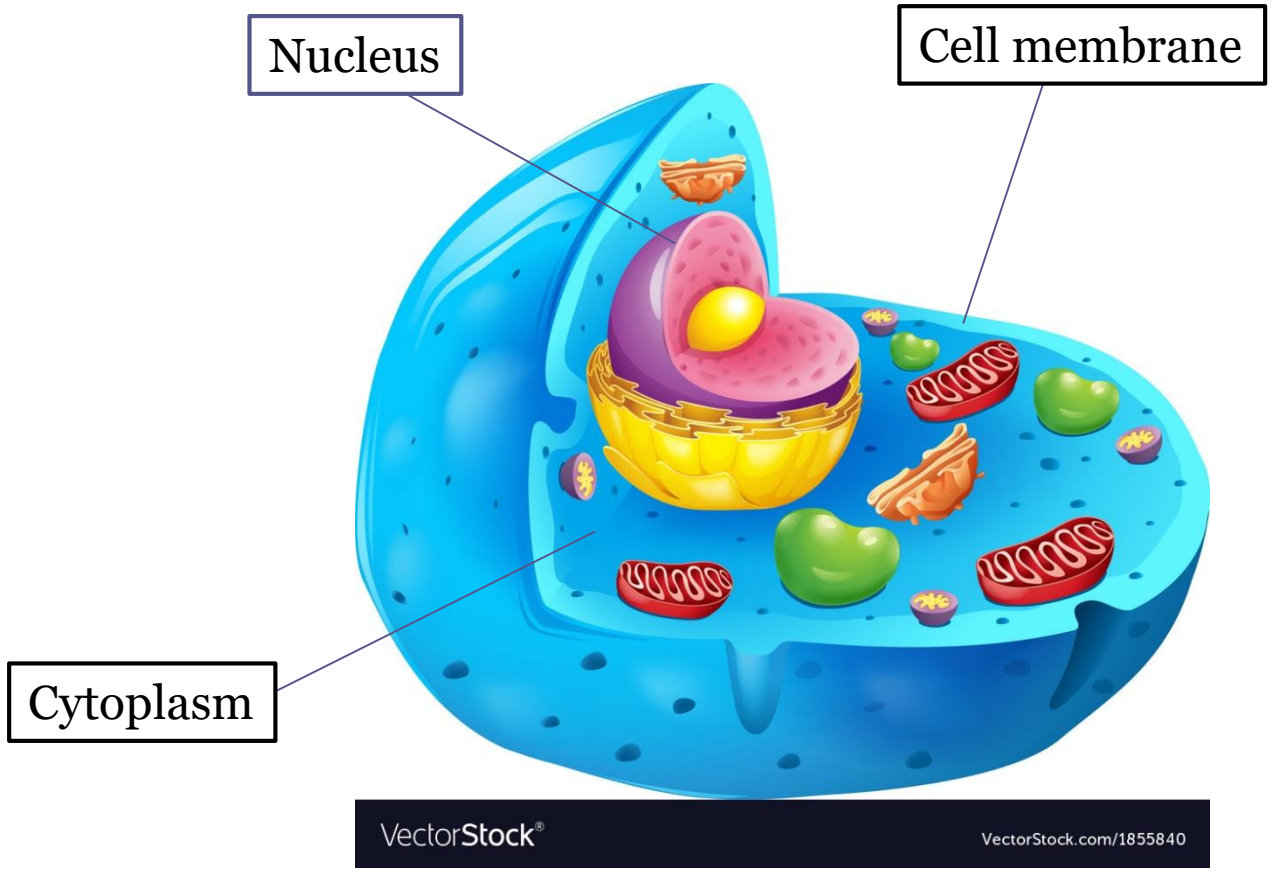


Protoplasm vs cytoplasm vs cytosol?

Protoplasm: Everything included in the cell membrane
divided into nucleoplasm vs cytoplasm

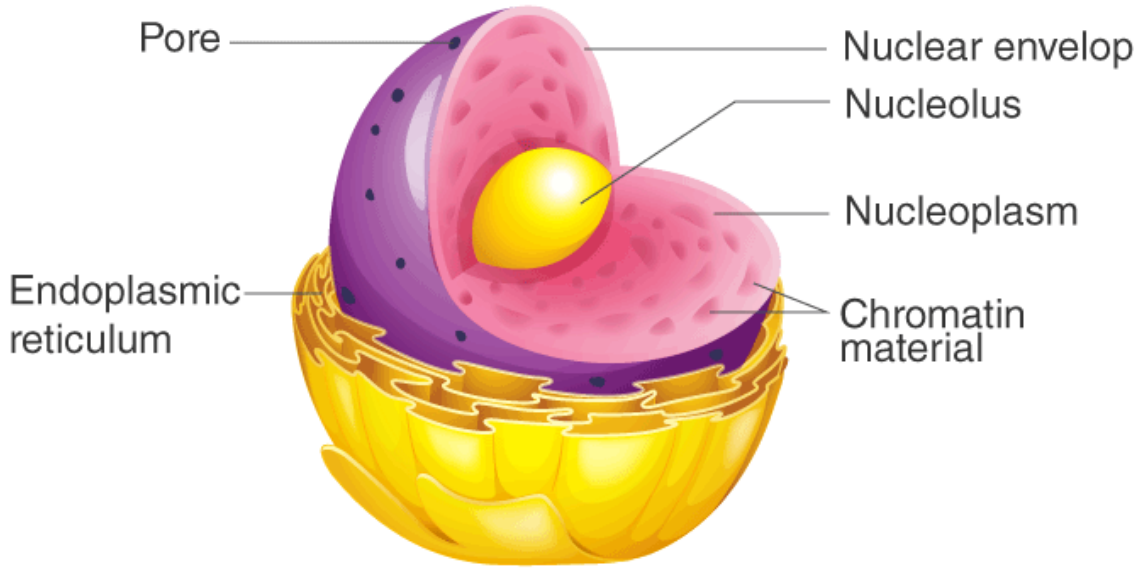
Cytoplasm: Everything included the cell membrane except the nucleus
Contains cytosol and organelles

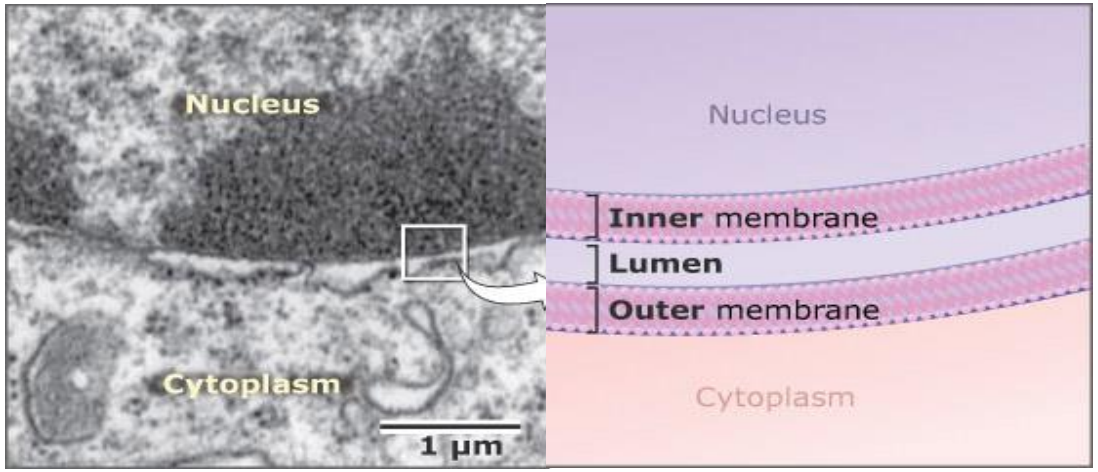
Cytosol: Area of cytoplasm that is not held by organelles
(cytoplasm minus the organelles)


Nucleus
Cell membrane
Cytoplasm
VectorStock®
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The nucleus

1. The **largest organelle** in the cell and is **bounded by an envelope** consisting of a **double membrane.**
2. **Genetic material** is concentrated in one part of the nucleus.
3. **Nuclear pores**: regulate the entry and exit of materials
4. **Nucleolus: ribosome biogenesis**

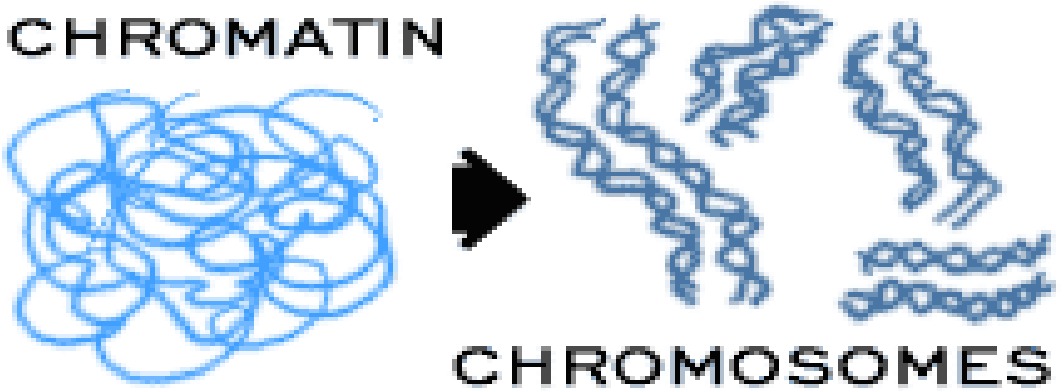




The nucleus is surrounded by the nuclear envelope, which consists of an inner membrane and an outer membrane. The membranes are separated by a lumen that is continuous with the lumen of the endoplasmic reticulum.

Inside the Nucleus

The genetic material (DNA) is found



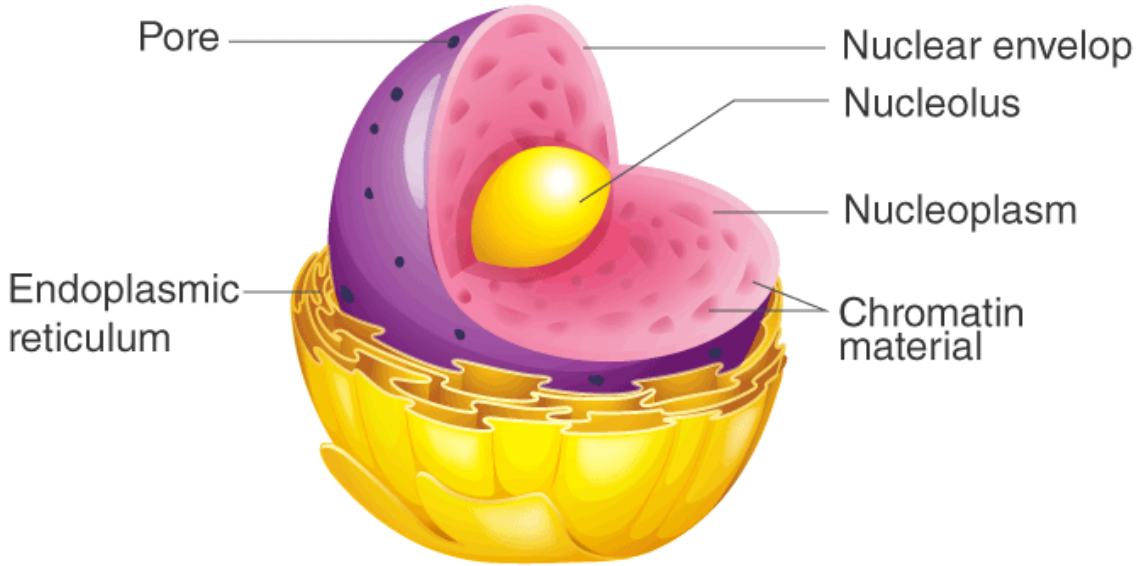
DNA is spread out and appears as chromatin in non-dividing cells

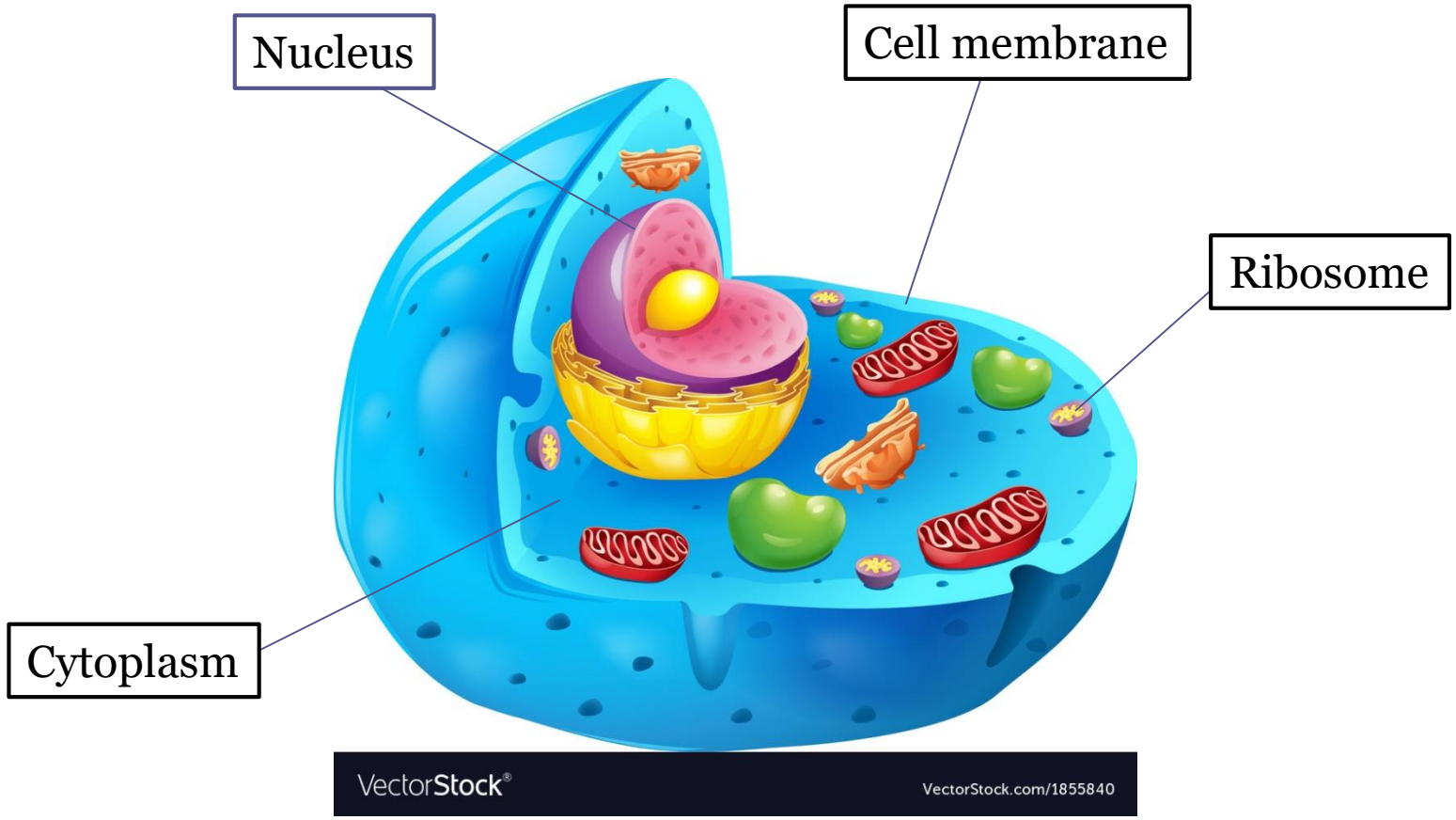
DNA is condensed & wrapped around proteins forming as chromosomes in dividing cells

The nucleus

Highlight the functions of the nucleus.

- It is responsible for storing the cell's hereditary material or the DNA.
- It is responsible for coordinating many of the important cellular activities such as protein synthesis, cell division, growth and a host of other important functions.




Nucleus
Cell membrane
Ribosome
Cytoplasm
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Ribosomes

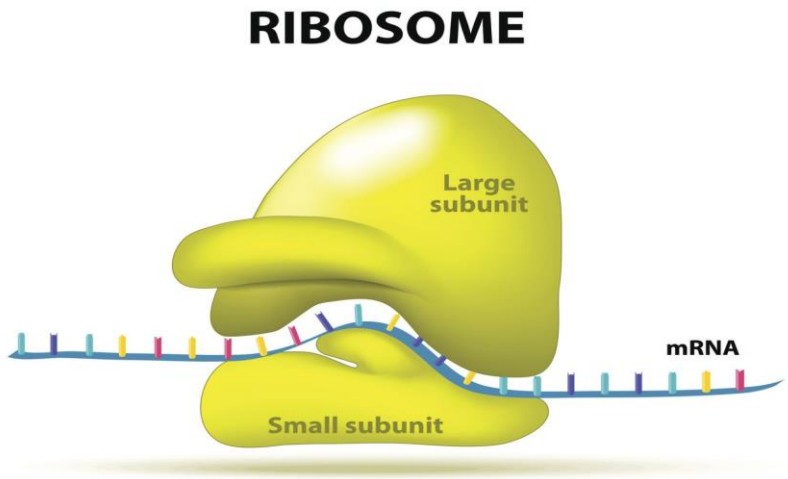
Made of proteins and rRNA

“Protein factories” for cell

(Join amino acids to make proteins through protein synthesis)

They are floating in the cytosol (make proteins that will be used inside)

or on ER (will be used inside and export)

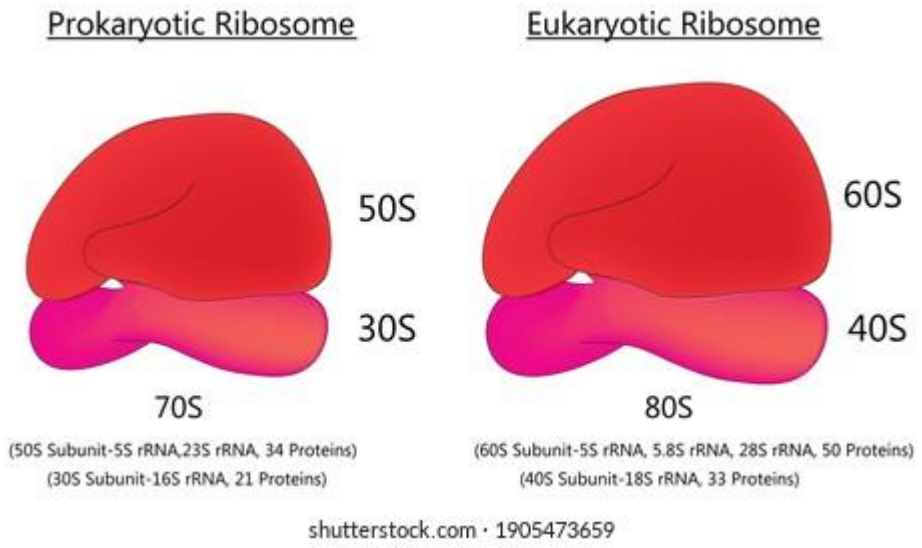


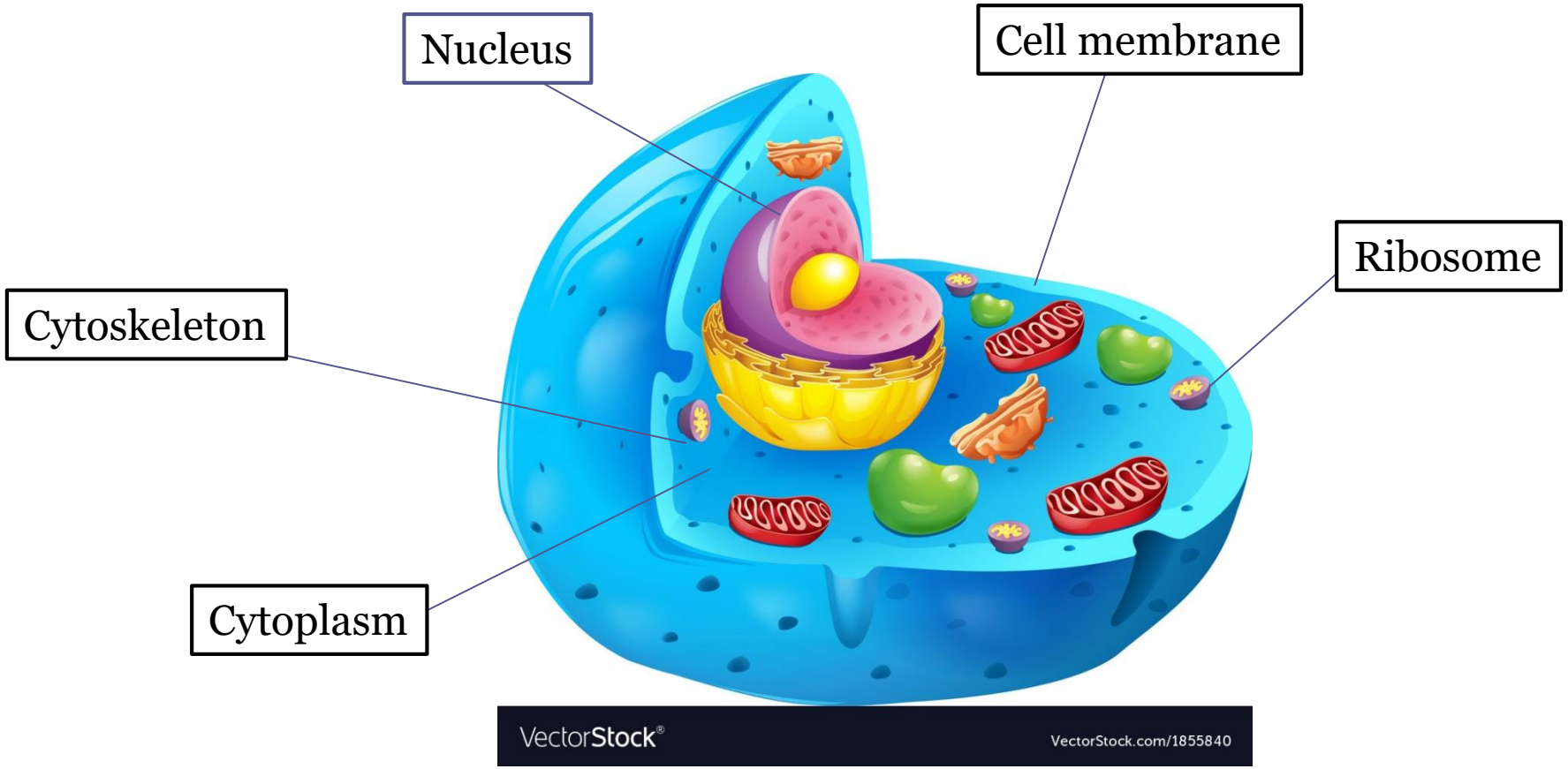
Ribosomes

They are found in both prokaryotes and eukaryotes

In eukaryotes: the 60-S (large) and 40-S (small) subunits → 80S

In Prokaryotes: 50-S and 30-S subunits → 70S




Nucleus
Cell membrane
Ribosome
Cytoskeleton
Cytoplasm
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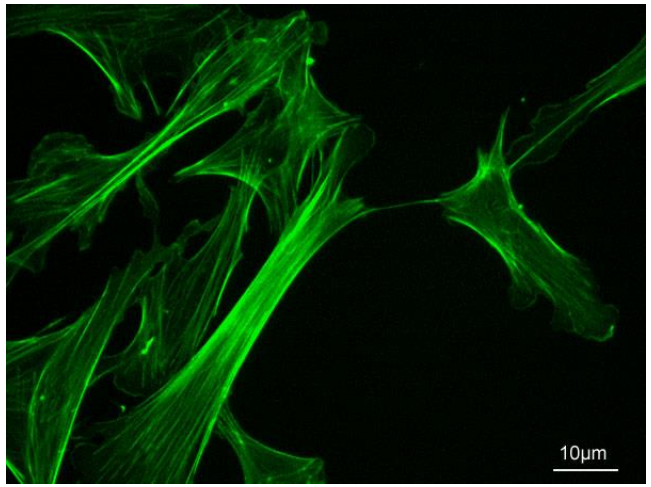
Cytoskeleton

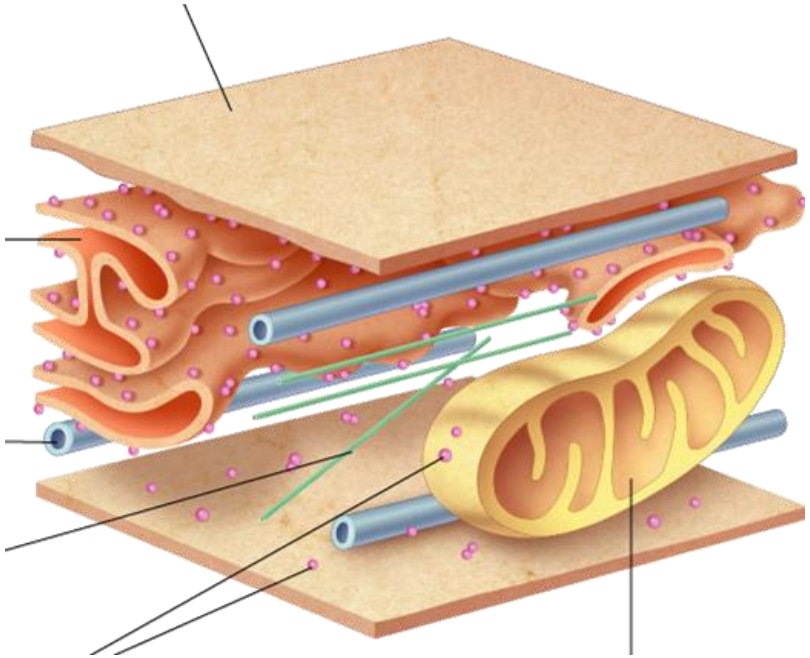
- Helps cells maintain cell shape
- Intracellular transport (the movement of vesicles and organelles within the cell) – extracellular transport
- Made of proteins

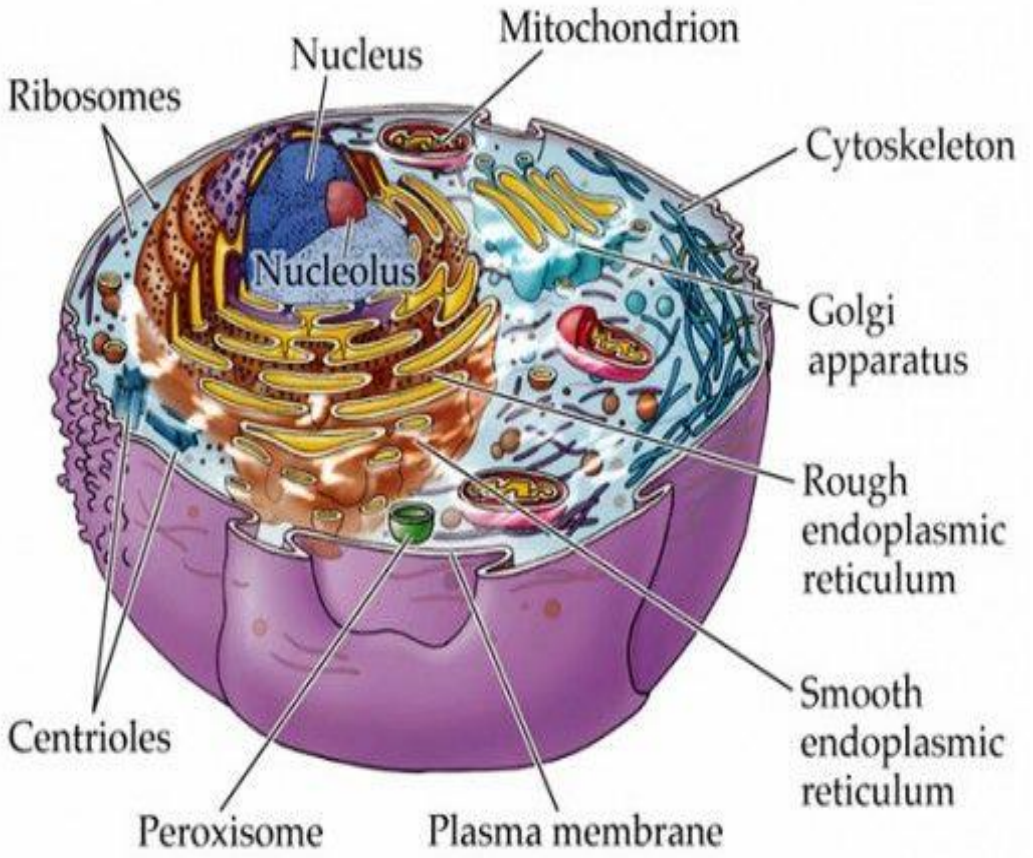
+ Microfilaments are threadlike and made of actin

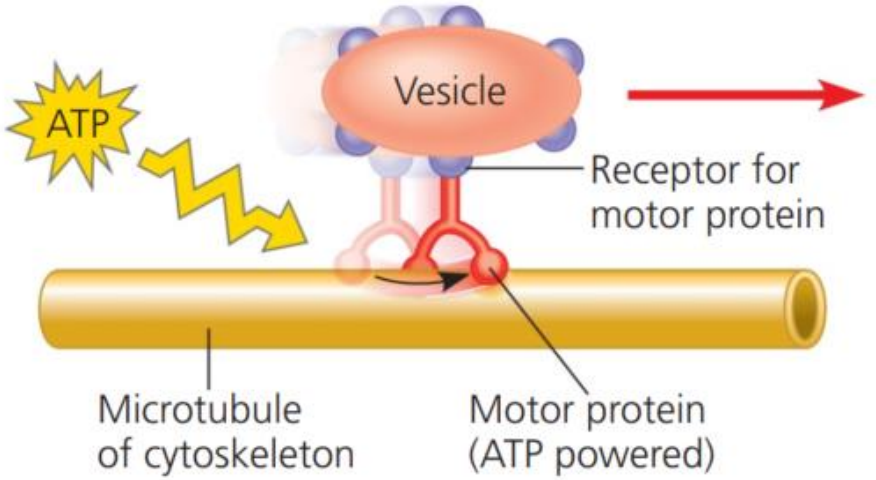
+ Microtubules are tube-like & made of tubulin

+ Intermediate filament









Motor proteins and the cytoskeleton

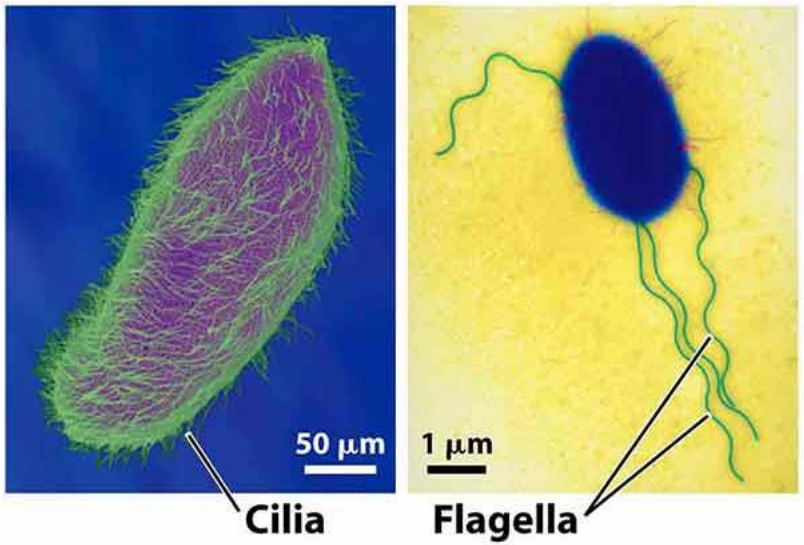
Motor proteins that attach to receptors on vesicles can "walk" the vesicles along microtubules or, in some cases, along microfilaments.

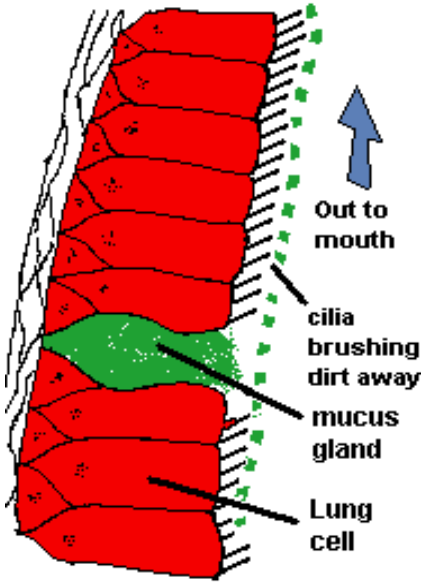
Cilia and flagella

Function in moving cells, in moving fluids, or in small particles across the cell surface

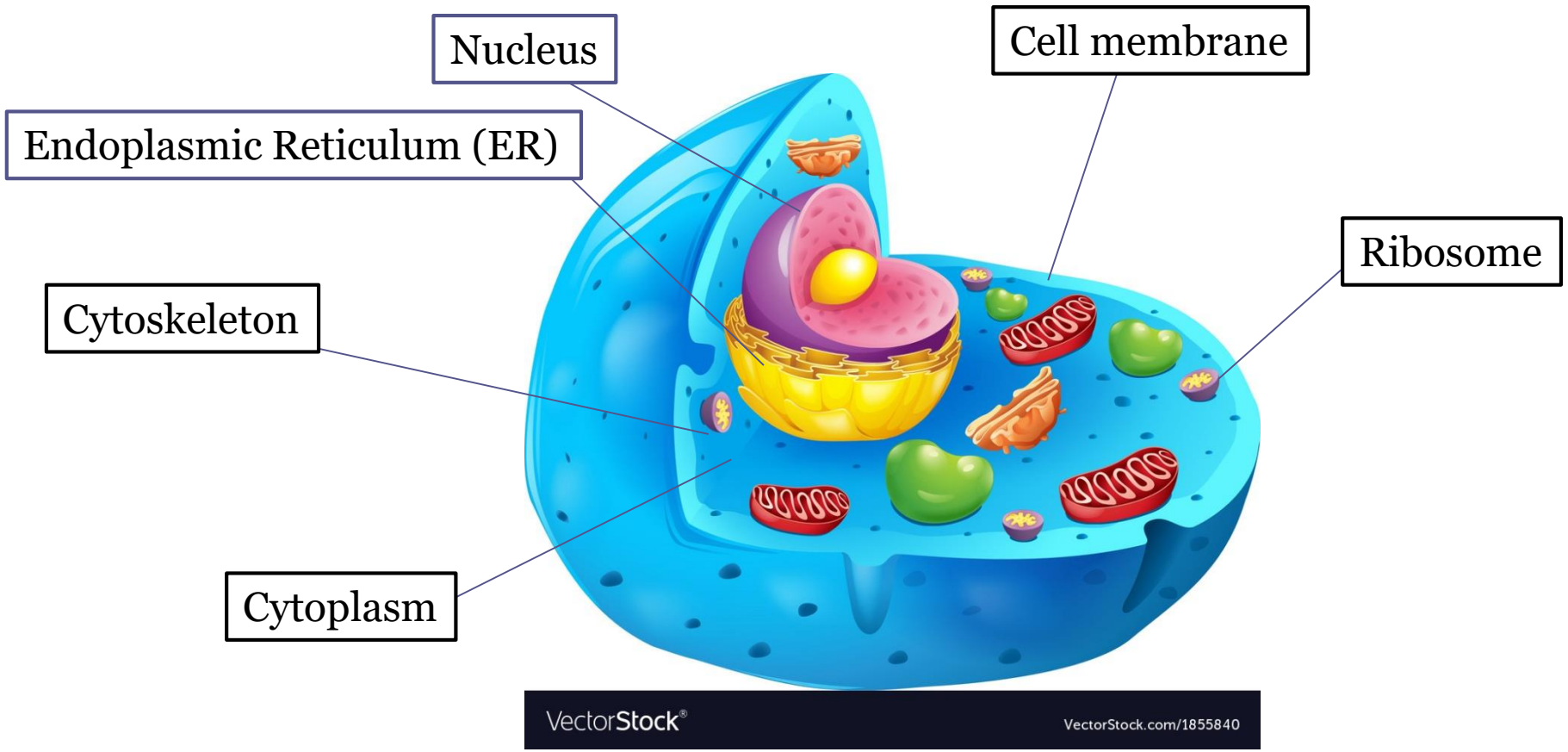
Cilia are shorter and more numerous on cells (Eukaryotic cells)

Flagella are longer and fewer (usually 1-3) on cells (Eukaryotic and prokaryotic cells)





Cilia Moving Away Dust Particles from the Lungs

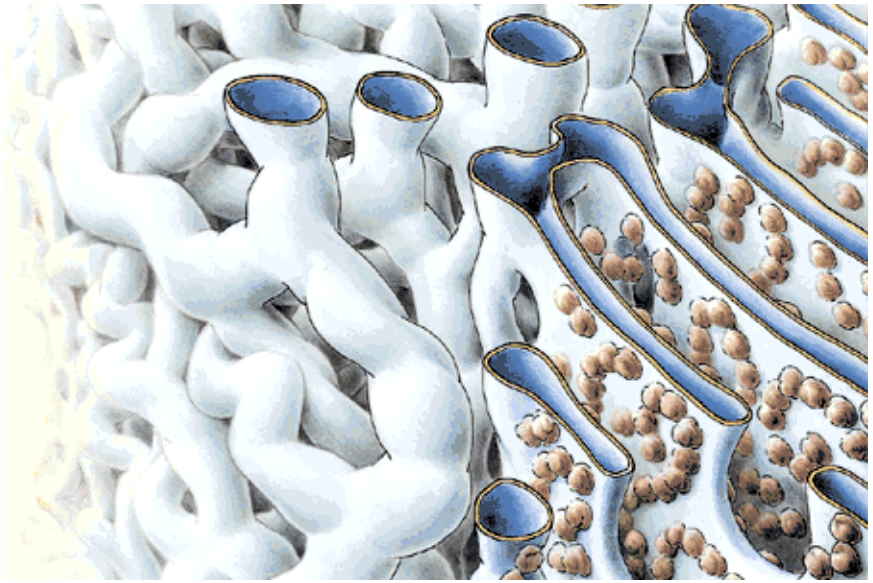

Nucleus
Cell membrane
Endoplasmic Reticulum (ER)
Ribosome
Cytoskeleton
Cytoplasm
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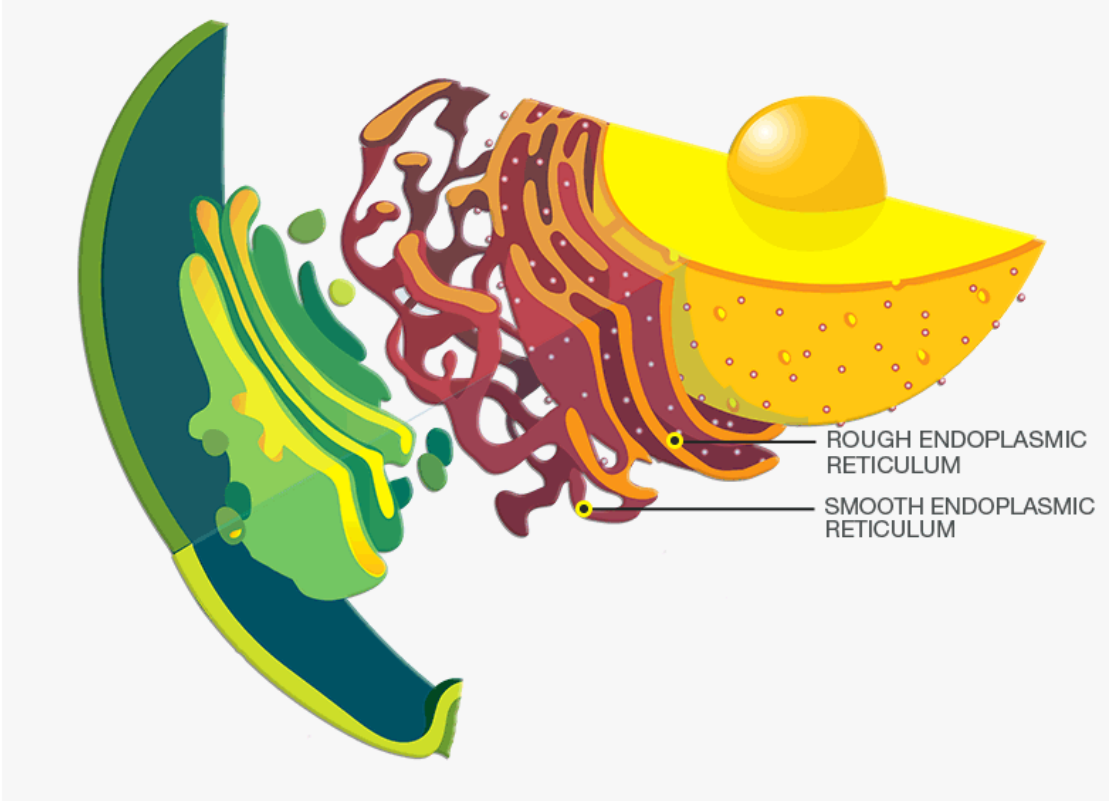
Endoplasmic Reticulum (ER)

Network of hollow membrane tubules

Connects to nuclear envelope and cell membrane

Functions in synthesis of cell products and Transport





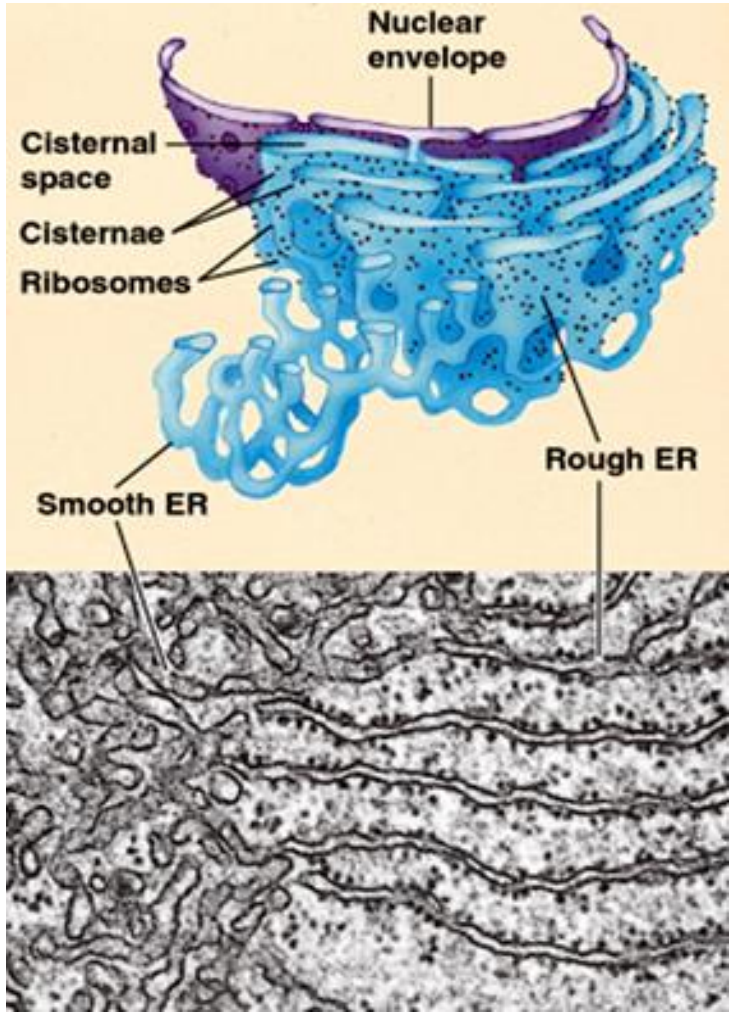
Two kinds of ER ---rough and Smooth

Rough (ER)

It is a series of connected flattened sacs having several ribosomes on its outer surface

Makes membrane proteins and proteins for export out of cell

Rough ER is prominent in cells where protein synthesis happens (such as hepatocytes)



Proteins are made by ribosomes on ER surface. They are then threaded into the interior of the Rough ER to be modified and transported

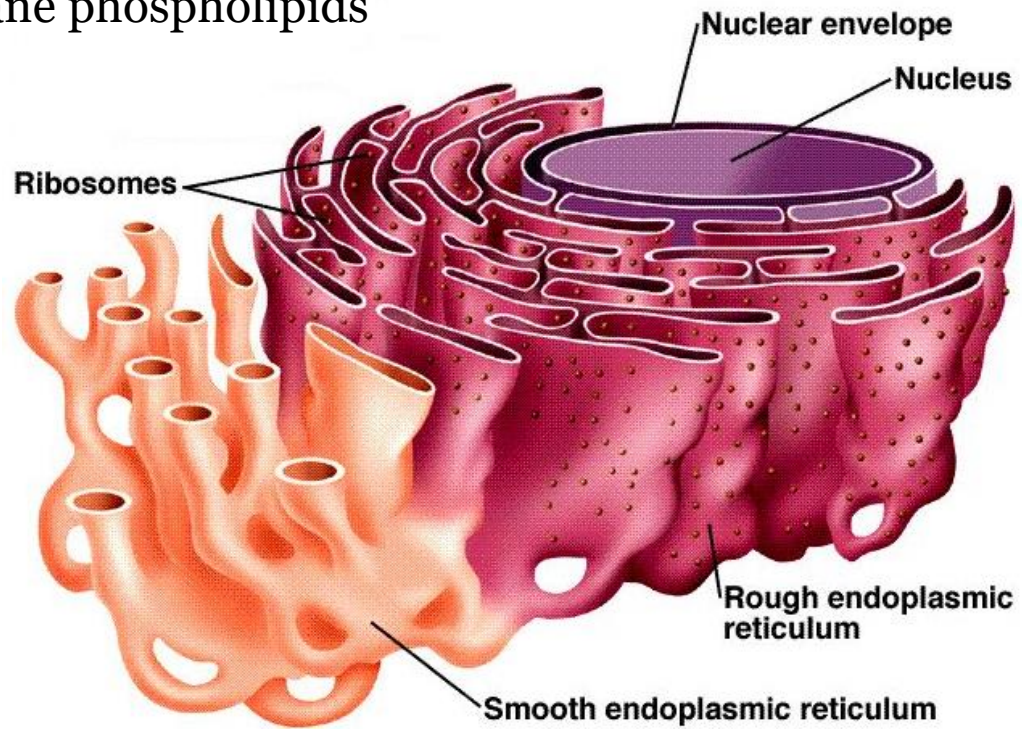
Smooth (ER)

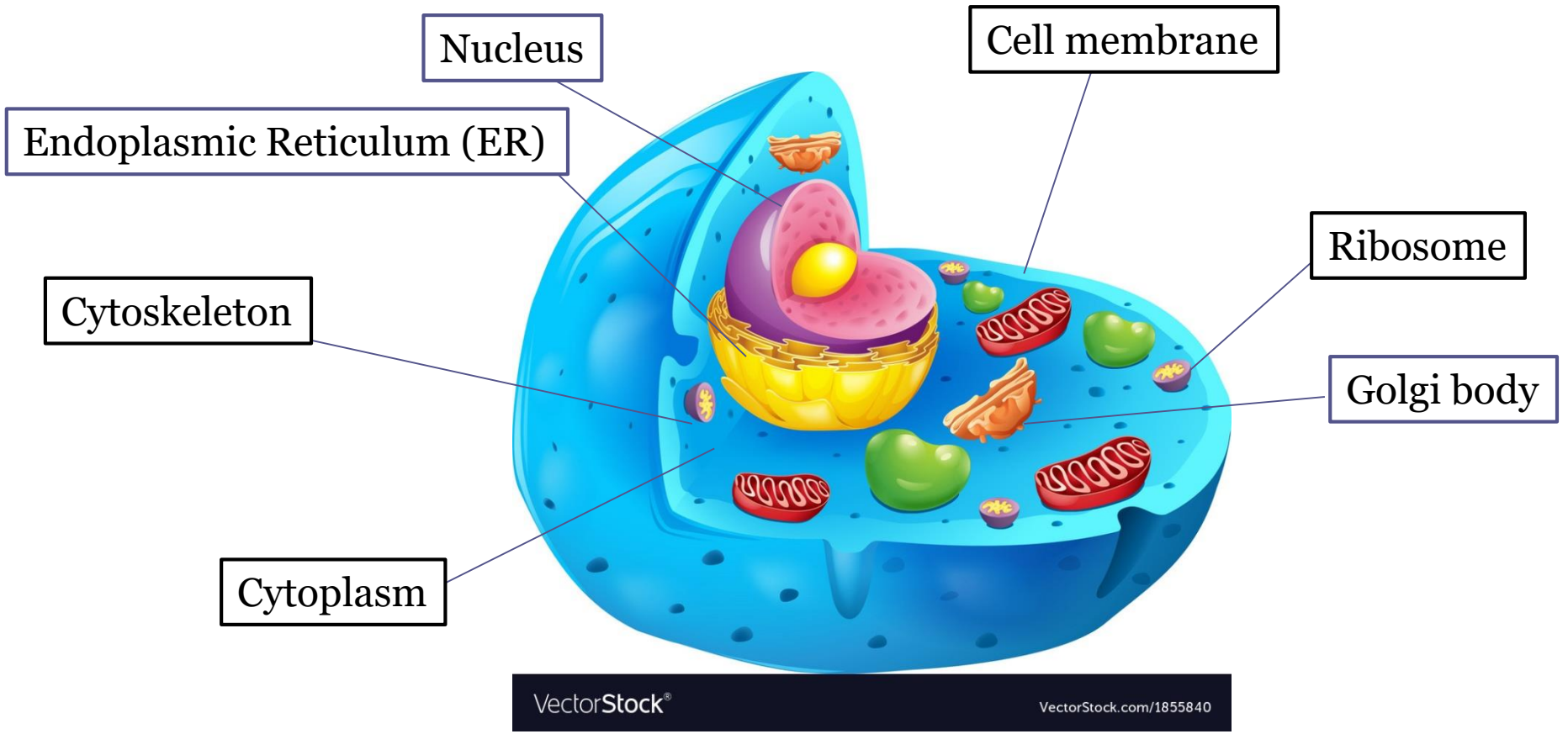
Has a tubular form

The smooth ER **stores and releases calcium ions**. These are quite important for the nervous system and muscular systems.

Synthesis of lipids, including oils, steroids, and new membrane phospholipids

Destroys toxic substances



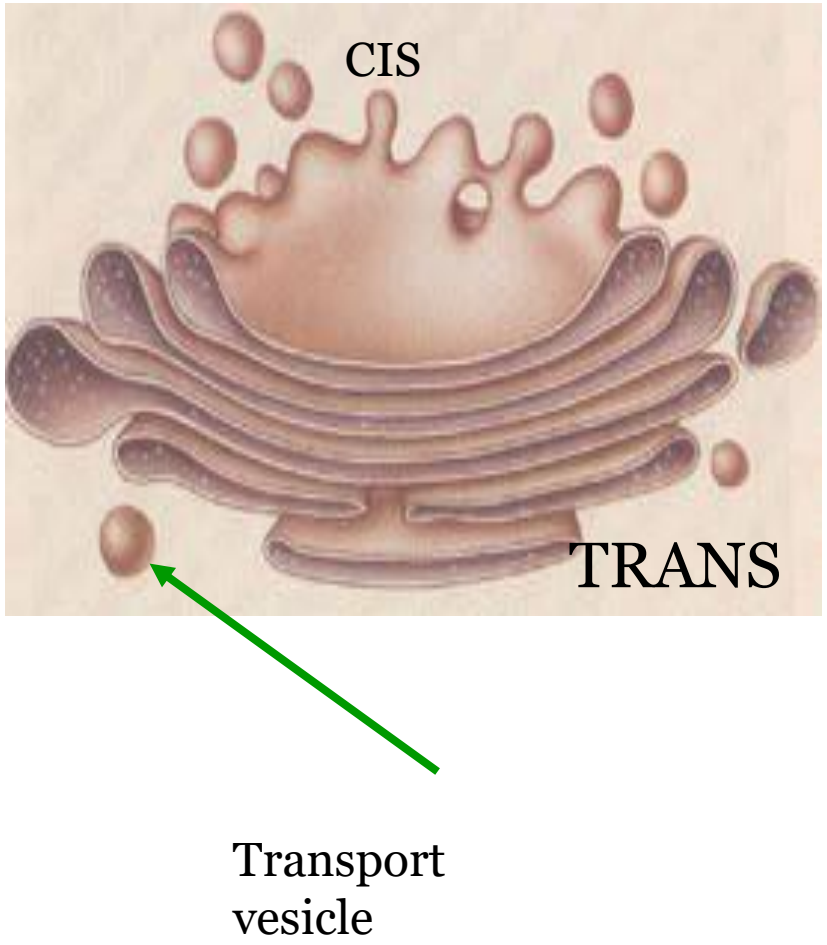

Nucleus
Cell membrane
Endoplasmic Reticulum (ER)
Ribosome
Cytoskeleton
Golgi body
Cytoplasm
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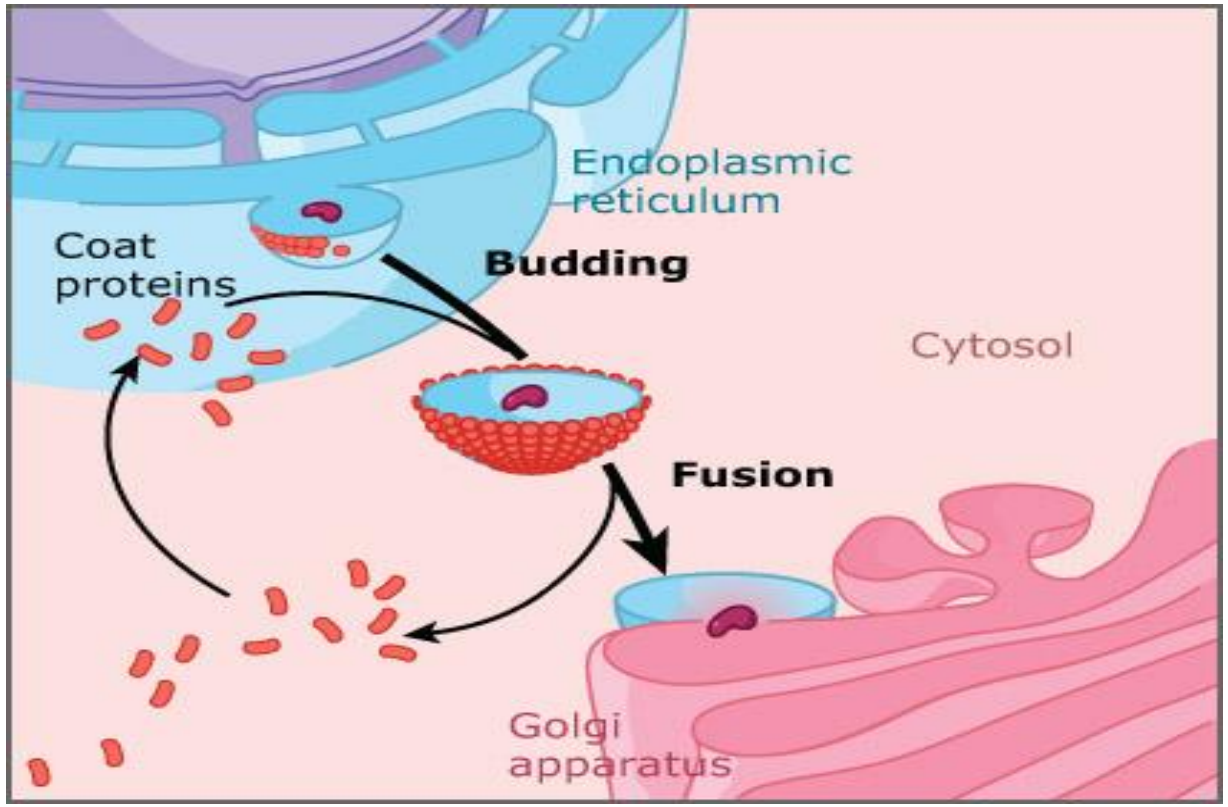
Golgi Body

- Stacks of flattened sacs
- Have a **receiving side (cis face)** and a **shipping side (trans face)**

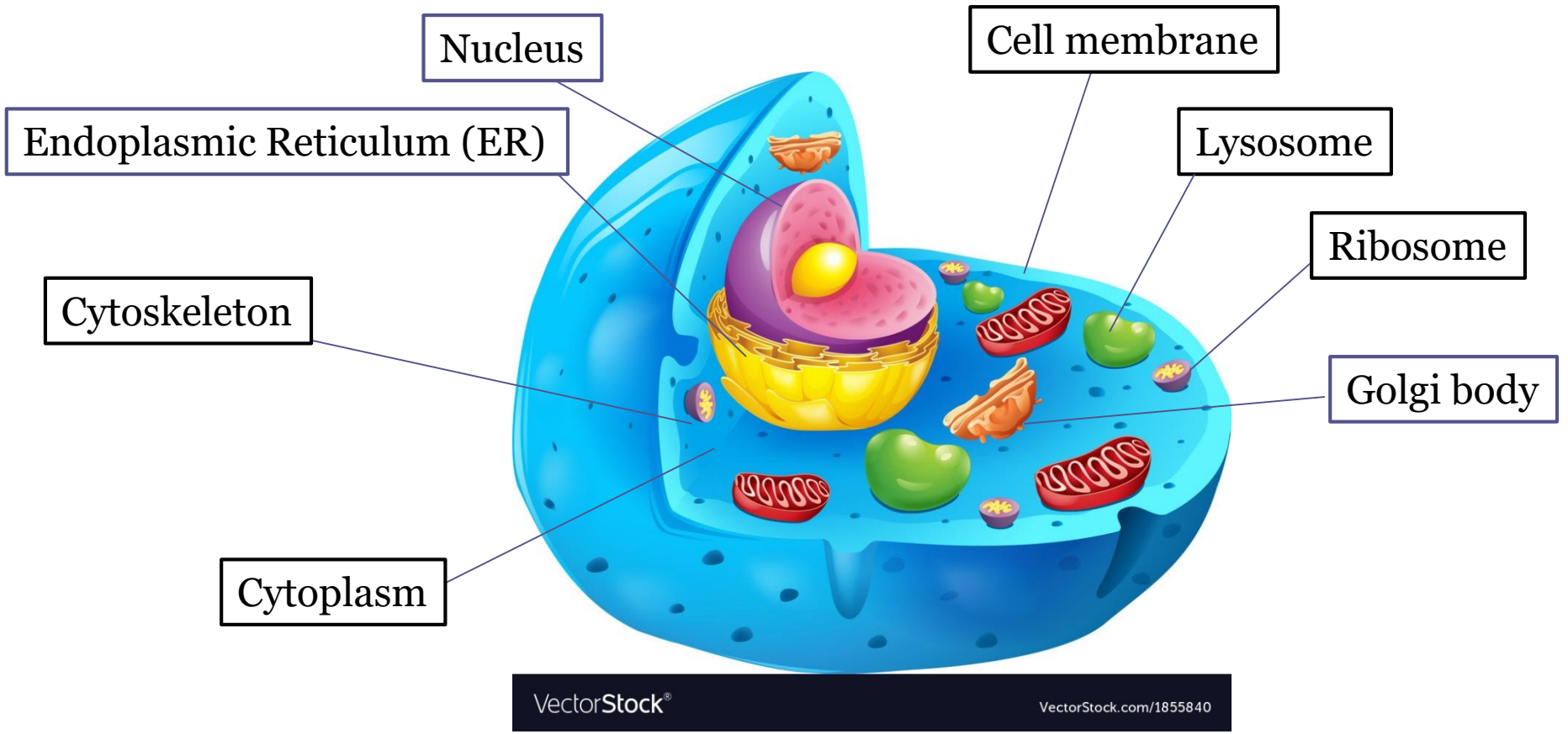
- > Receive proteins made by ER by cis face, *trans* face secretes the materials into vesicles, which then fuse with the cell membrane for release from the cell

In the plant cells, complex polysaccharides of the cell wall are synthesized in the Golgi apparatus.



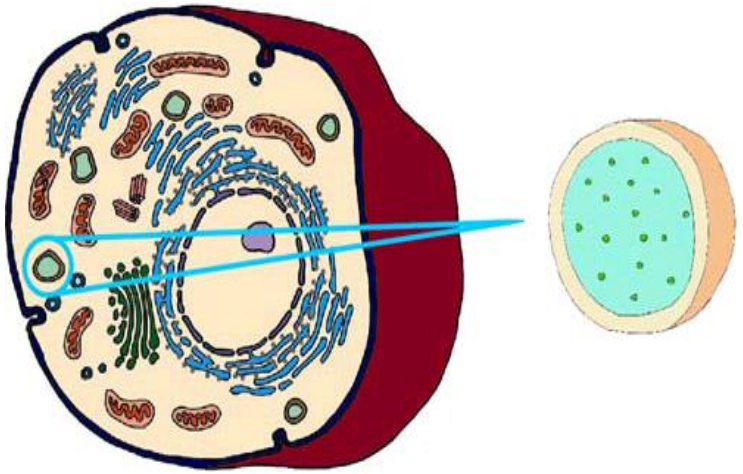


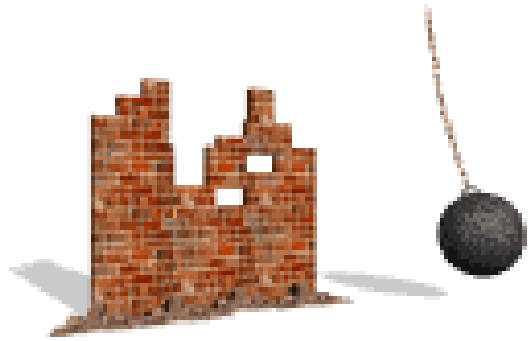
Transport of proteins between membrane-bounded compartments occur when vesicles containing the proteins bud from one compartment and subsequently fuse with another compartment


Nucleus
Cell membrane
Endoplasmic Reticulum (ER)
Lysosome
Ribosome
Cytoskeleton
Golgi body
Cytoplasm
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Lysosome

- Contain digestive enzymes
- Break down food, bacteria, and worn out cell parts for cells
- Lyse and release enzymes to break down and recycle cell parts





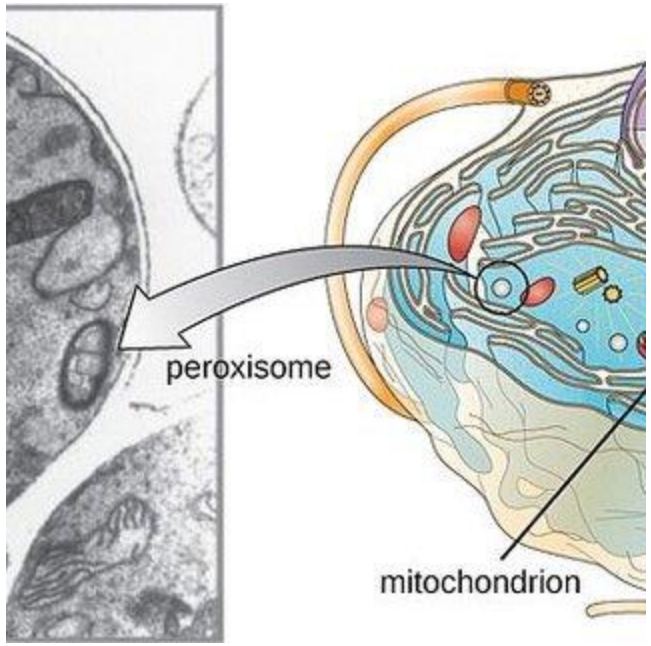
Little Enzyme Packages

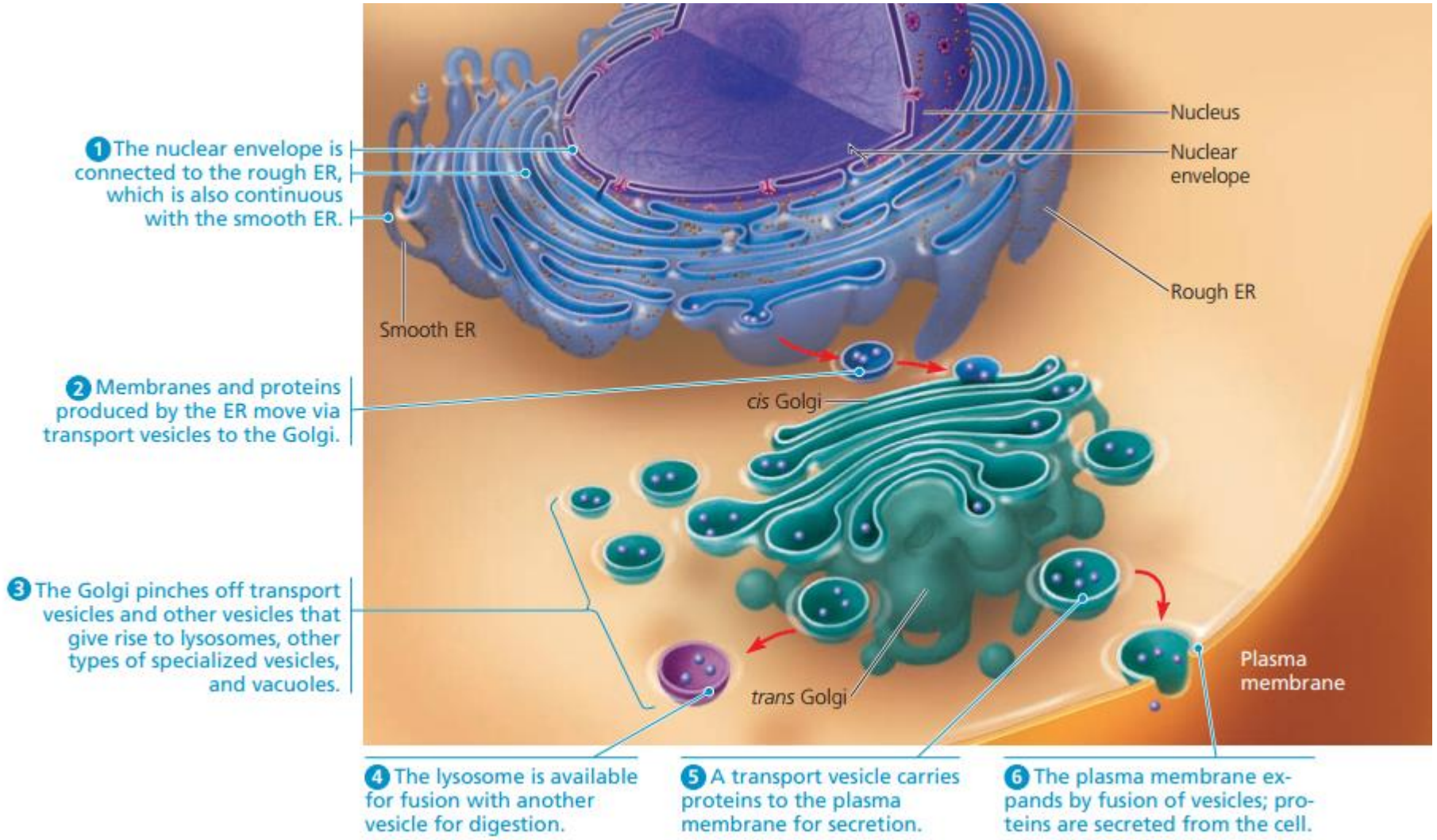
Peroxisome -Another Enzyme Package

• Small vesicles found around the cell

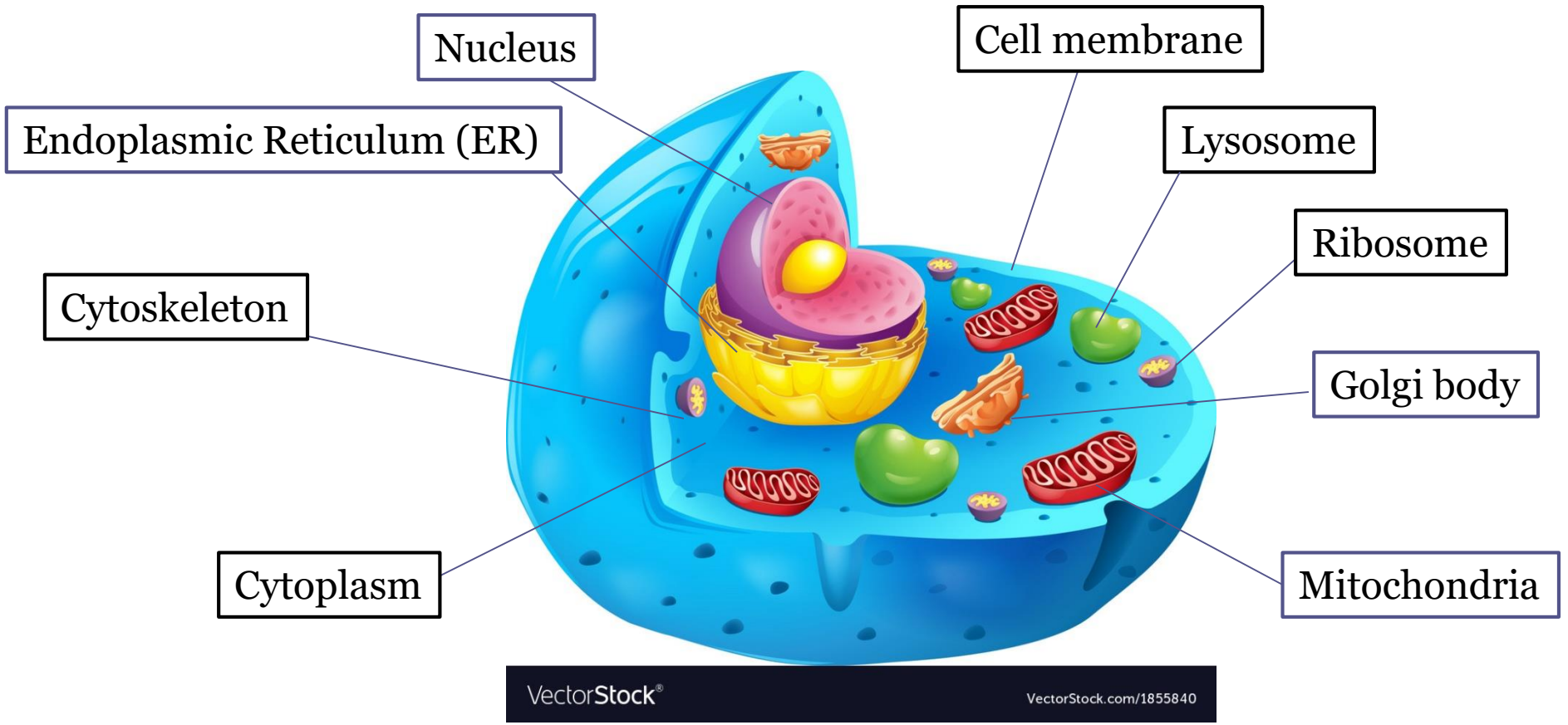
•Contains digestive enzymes (enzymes that require oxygen (oxidative enzymes))

Absorb nutrients that the cell has acquired, digesting **fatty acids,** digest **alcohol,** cholesterol synthesis and the digestion of **amino acids**





Review: relationships among organelles of the endomembrane system.
The red arrows show some of the migration pathways for membranes and the materials they enclose


Nucleus
Cell membrane
Endoplasmic Reticulum (ER)
Lysosome
Ribosome
Cytoskeleton
Golgi body
Cytoplasm
Mitochondria
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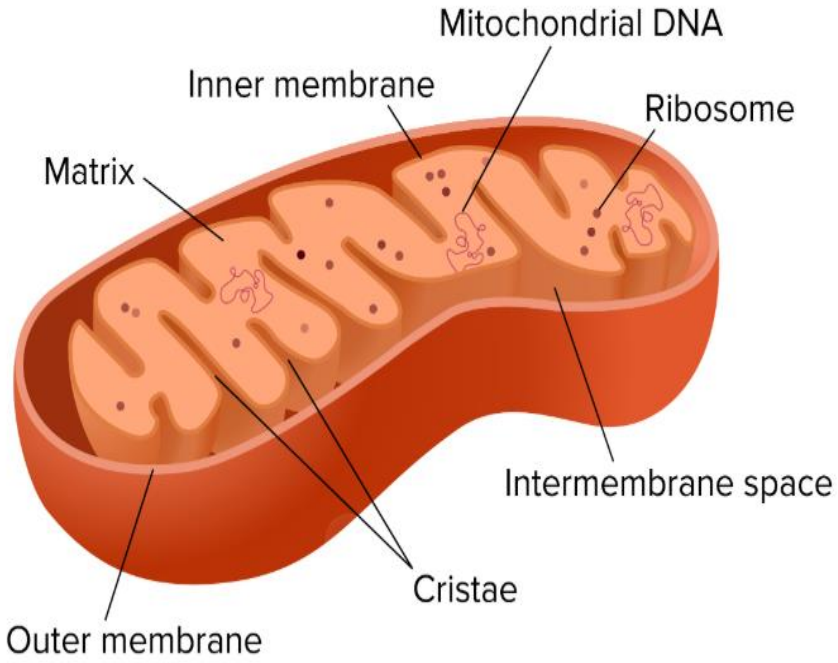
Mitochondria are energy factories

Key concept

All living cells have a means of converting energy supplied by the environment into the common intermediate of ATP

❖ Structure:

- rod-shaped structure,
- a double-membraned bound, **outer membrane** smooth, **inner membrane** folded to form a structure called **cristae.**
- The fluid contained in the mitochondria is called the **matrix:** contains a mixture of enzymes and proteins. It also comprises ribosomes, inorganic ions, mitochondrial DNA, nucleotide cofactors, and organic molecules.



Mitochondria are energy factories

Functions

The most important function: converts energy stored in food into usable energy for work – **cellular respiration/** (burning glucose)/ power house of the cell/ (Generate cellular energy (ATP))

Others:

Regulates the metabolic activity of the cell

Promotes the growth of new cells and cell multiplication

Helps in detoxifying ammonia in the liver cells

Plays an important role in apoptosis or programmed cell death

Is the Mitochondria genome still functional?

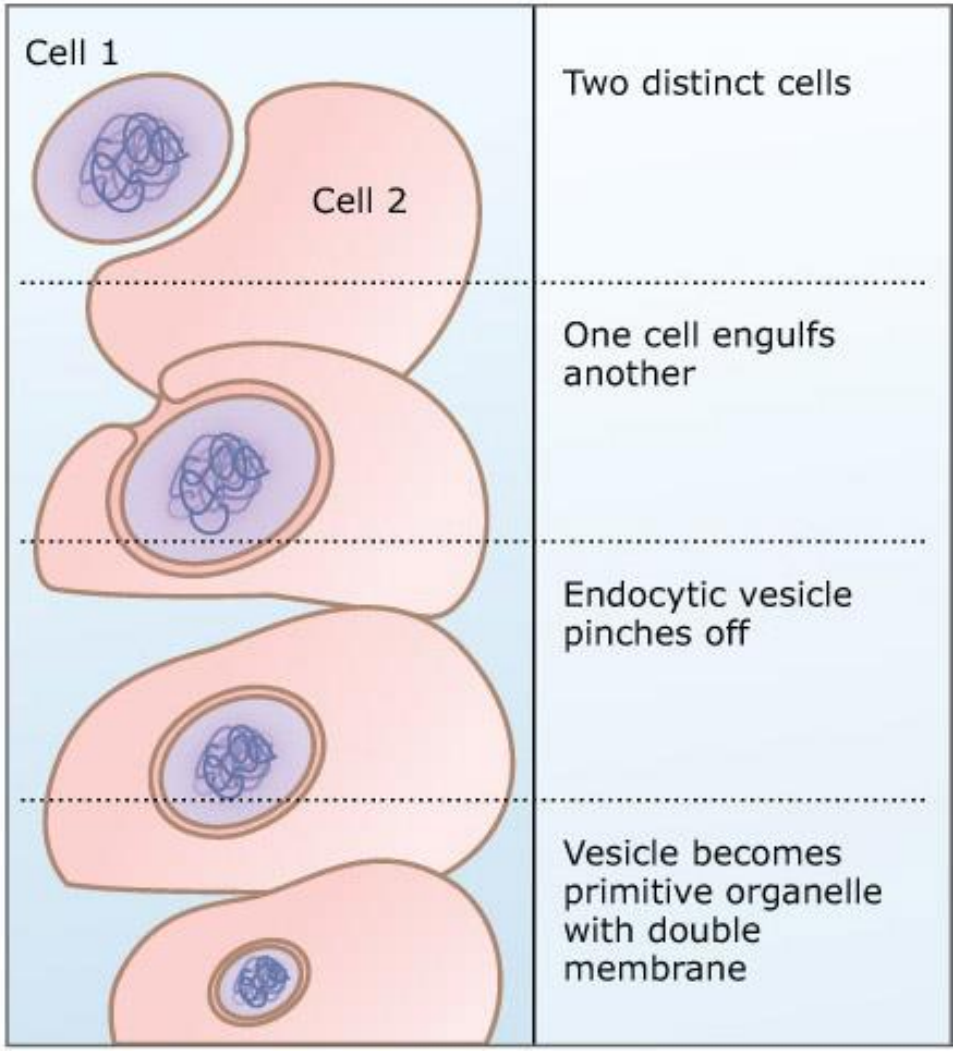
Mitochondria are special because they have **their own ribosomes** and **DNA** floating in the matrix

Have been conserved across evolution

Retains similarity to its prokaryotic ancestor

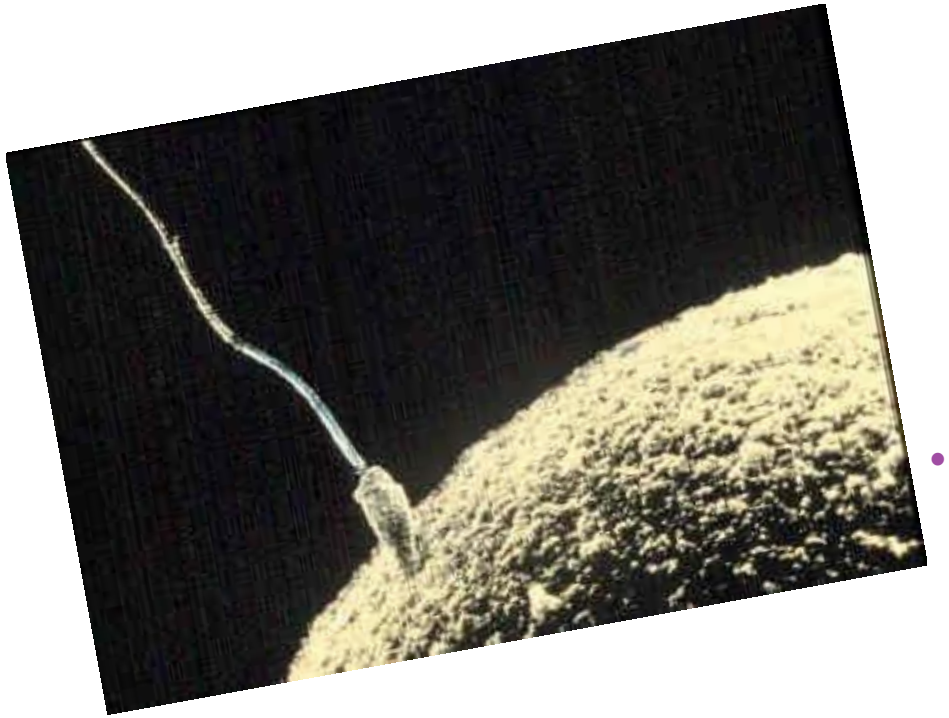
Can you guess the origin of Mitochondria?

Mitochondria are thought to have originated from an ancient symbiosis that resulted when a nucleated cell engulfed an aerobic prokaryote



Enveloped organelles could have evolved when one cell ingested another.

Interesting Fact ---



- Mitochondria come from cytoplasm in the EGG cell during fertilization

Therefore …

- You inherit your mitochondria from your mother!

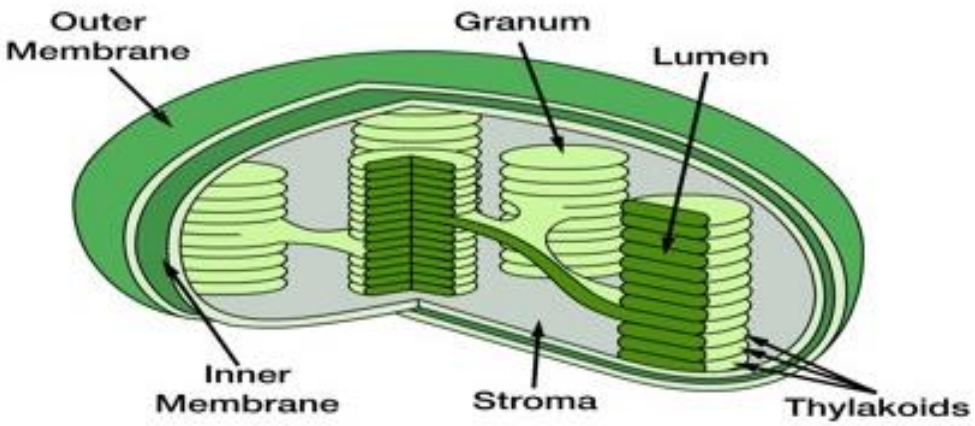


Chloroplasts: power plant cells

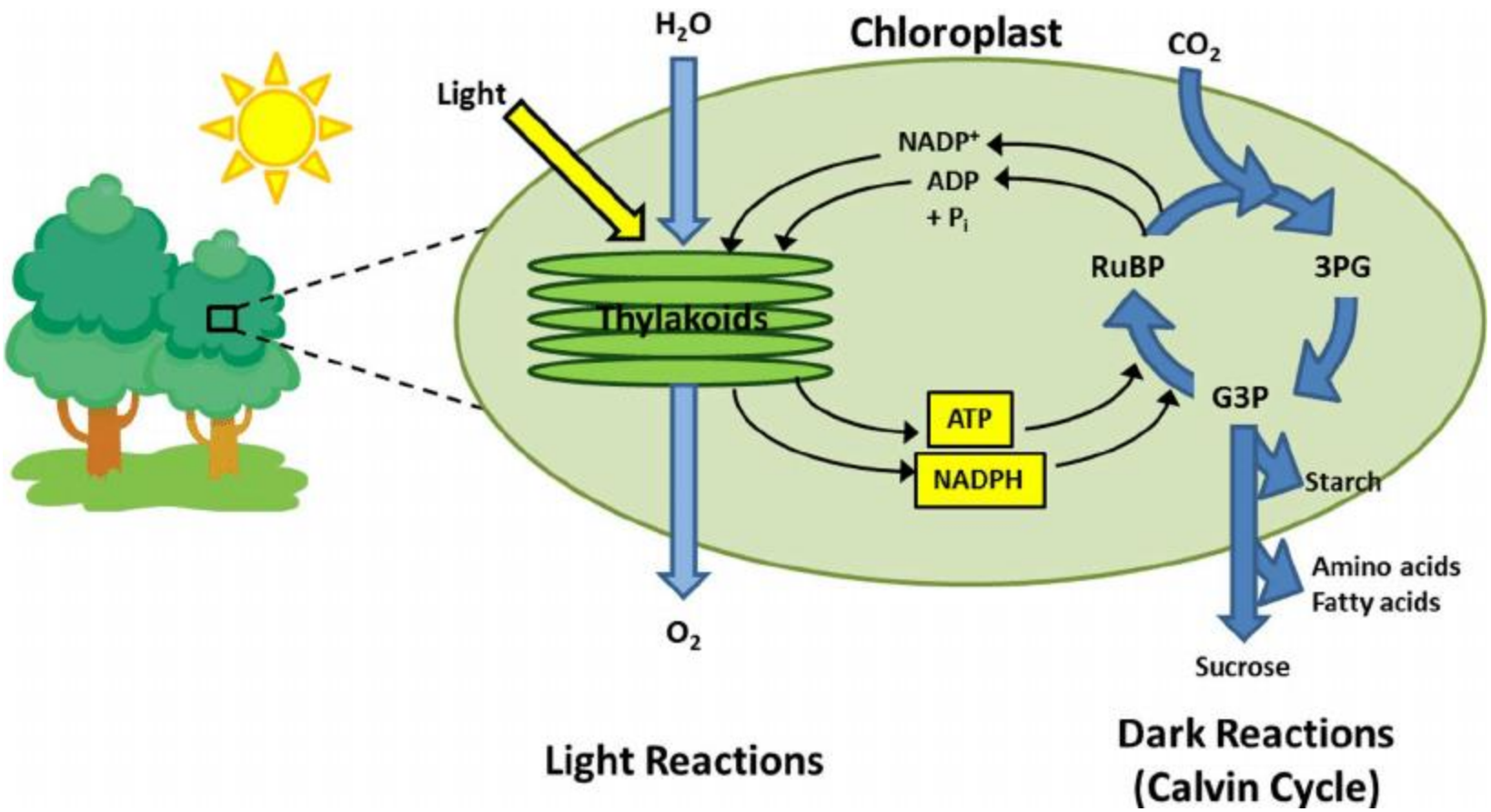
Chloroplasts work to convert the light energy of the Sun into sugars that can be used by cells - **photosynthesis**

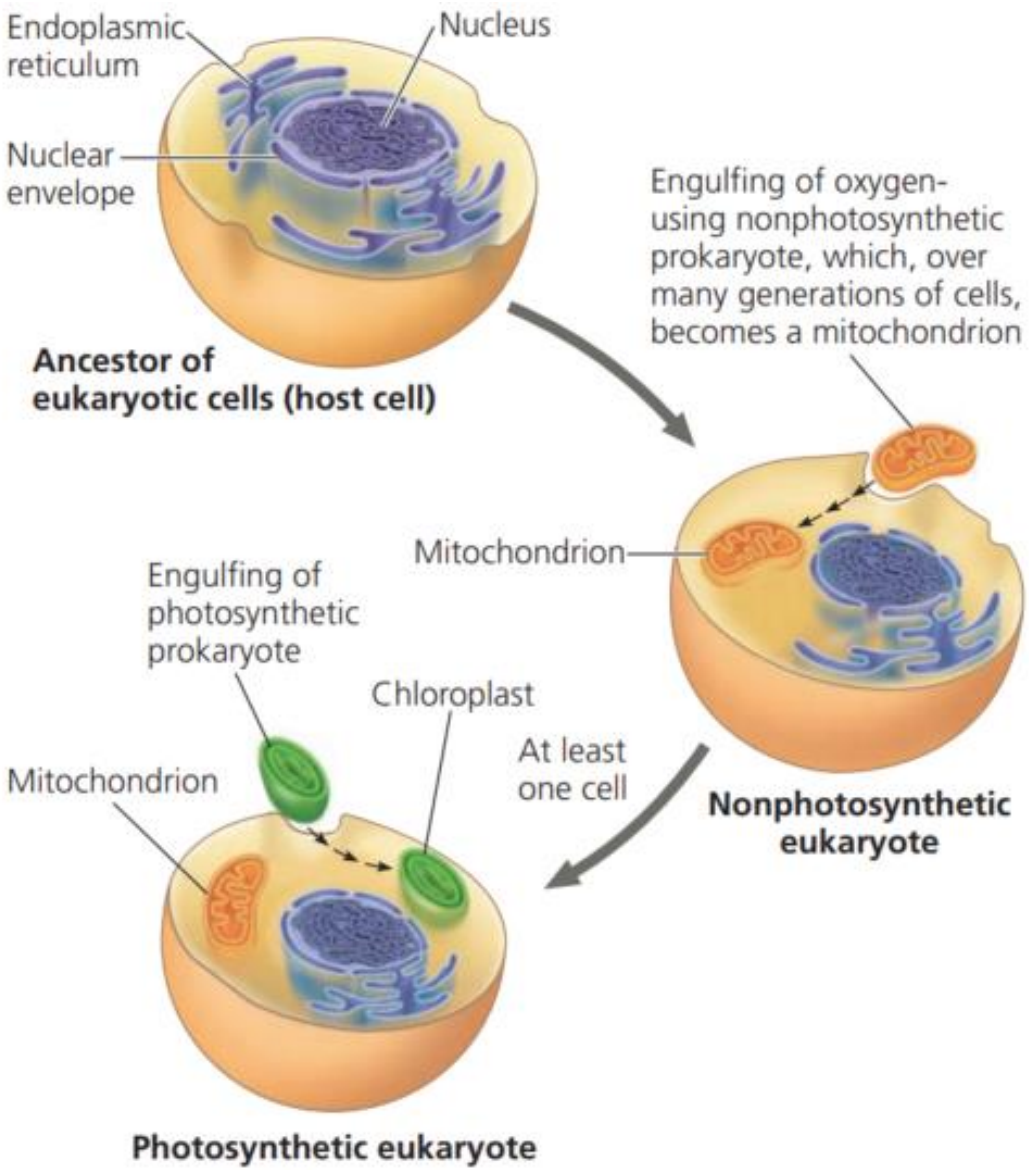
Found in plant cells and some protists such as algae

- Outer membrane smooth
- Inner membrane modified into sacs called Thylakoids
- Thylakoids in stacks called Grana and interconnected
- Stroma – gel like material surrounding thylakoids



Photosynthesis





The endosymbiont theory of the origins of mitochondria and chloroplasts in eukaryotic cells.

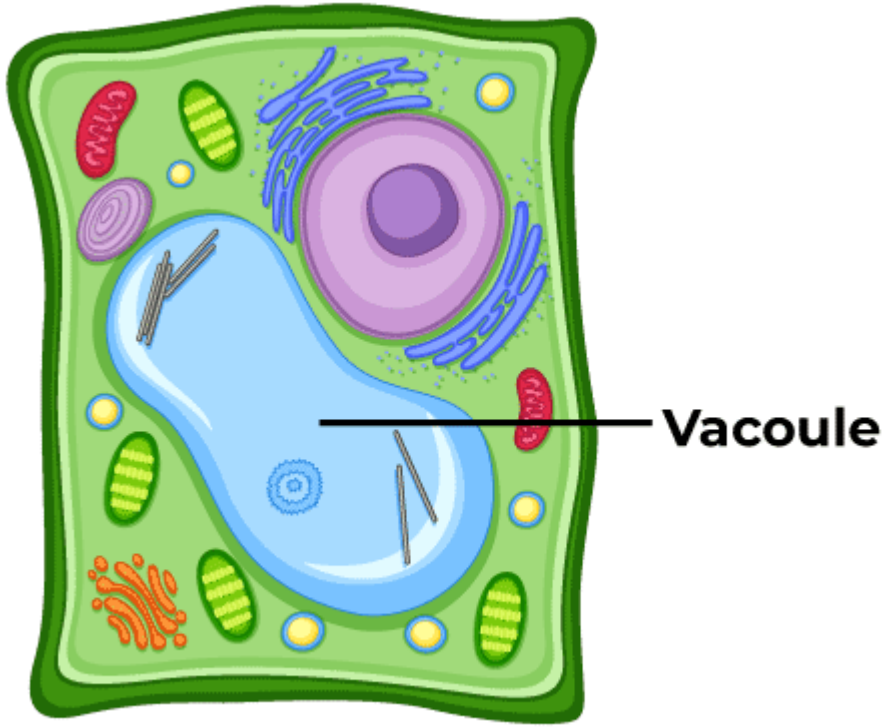
The proposed ancestors of mitochondria were oxygen-using nonphotosynthetic prokaryotes, while the proposed ancestors of chloroplasts were photosynthetic prokaryotes. The large arrows represent change over evolutionary time; the small arrows inside the cells show the process of the endosymbiont becoming an organelle, also over long periods of time.

Vacuoles

Fluid filled sacks for storage

Small or absent in *animal* cells

Plant cells have a large Central Vacuole

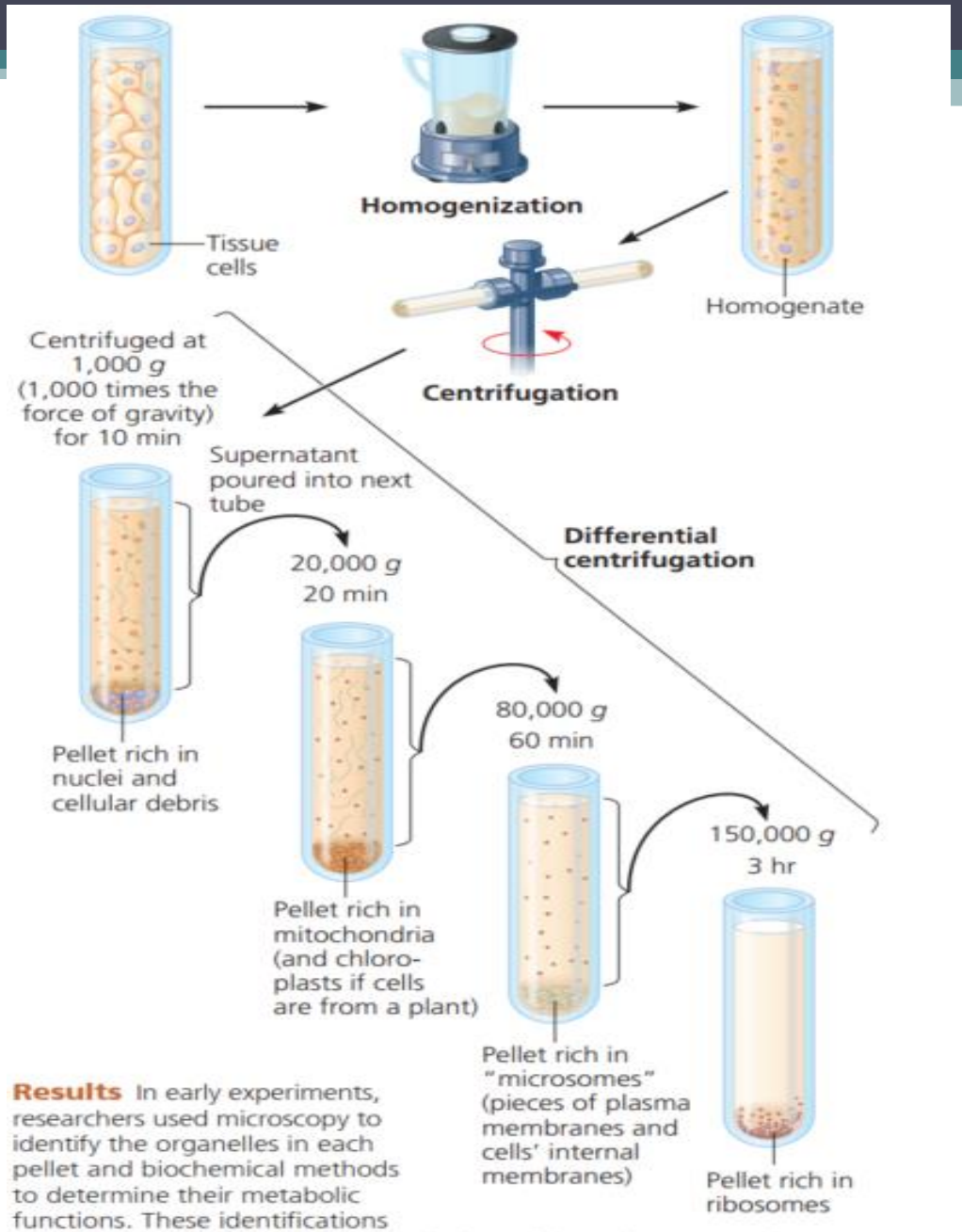


Functions of Vacuole:

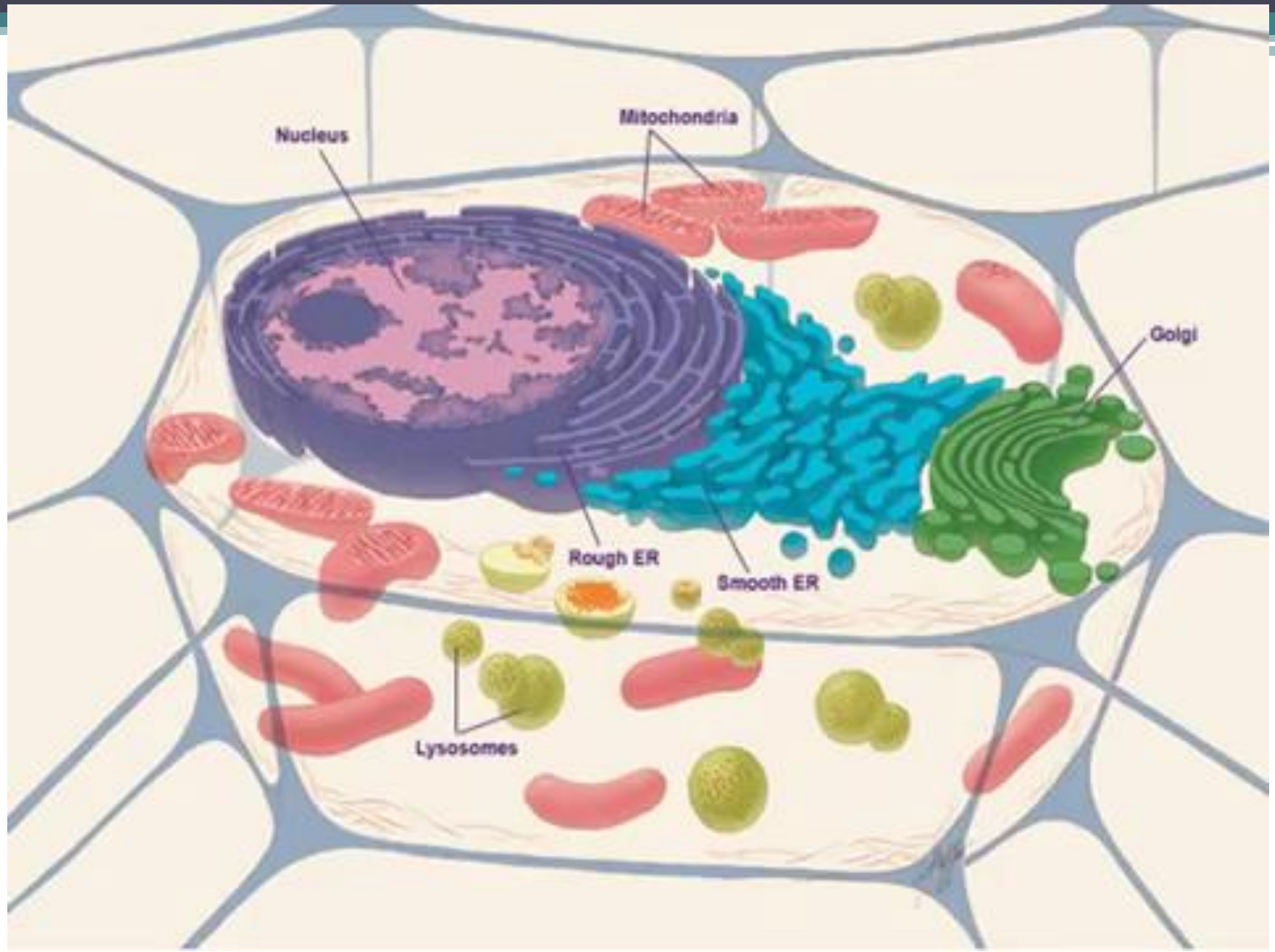
Vacuole helps in storage of salts, minerals, pigments and proteins within the cell.

It isolates metabolic waste that might be harmful to the cell.

It maintains turgor pressure.



Results In early experiments, researchers used microscopy to identify the organelles in each pellet and biochemical methods to determine their metabolic functions. These identifications established a baseline for this method, enabling today's researchers to know which cell fraction they should collect in order to isolate and study particular organelles.


Nucleus
Mitochondria
Golgi
Rough ER
Smooth ER
Lysosomes